

A Gym Management, Workout and Meal Plan Recommendation System using Optimization And Machine Learning

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TABLE OF CONTENT

CHAPTER I - THE PROBLEM AND ITS BACKGROUND	4
1.1 Project Context	4
Background of the Study.....	7
1.2 Statement of the Problem	9
1.3 Objective of the Study.....	9
General Objectives.....	9
Specific Objectives.....	10
1.4 Scope and Delimitation of the Study.....	10
Scope.....	11
Delimitations.....	12
1.5 Significance of the Study.....	13
1.6 Definition of Terms.....	14
CHAPTER II - REVIEW OF RELATED LITERATURE AND STUDIES.....	16
2.1 Introduction.....	16
2.2 Review of Related Literature.....	16
Foreign Literature.....	16
Local Literature.....	21
2.3 Review of Related Studies.....	27
Foreign Studies.....	27
Local Studies	35
2.4 Synthesis of the Review of Related Literatures and Studies.....	43
CHAPTER III - RESEARCH METHODOLOGY	45

3.1 Research Method.....	45
DFD (Data Flow Diagram) Level 0 Proposed System.....	47
DFD (Data Flow Diagram) Level 1 Proposed System	49
DFD (Data Flow Diagram) Level 2 Proposed System	50
3.2 Conceptual Framework of the Study.....	57
3.3 Data Gathering Tools and Procedure.....	59
Process Flowchart Existing System.....	62
3.4 Project Design.....	63
System Architecture Diagram.....	63
Use Case Diagram.....	66
Entity Relationship Diagram (ERD).....	68
Data Dictionary Table.....	69
3.5 Project Evaluation.....	76
Project Evaluation Instruments.....	76
AI Model Evaluation Metrics.....	79
Likert Table.....	81
Reference.....	82

CHAPTER I

THE PROBLEM AND ITS BACKGROUND

1.1 Background of the Study

This study proposes the development of an integrated Gym Management and Recommendation System powered by Machine Learning and Optimization Algorithms. The primary purpose of this system is to streamline core gym business processes while enhancing the member experience through intelligent automation. By combining administrative management functions with AI-driven insights, the system aims to generate personalized workout plans and optimally assign trainers based on predefined constraints. Ultimately, this solution seeks to bridge the gap between operational efficiency and personalized service, allowing gym establishments to scale their services while maintaining high standards of quality.

The proposed system focuses on improving several essential business processes commonly performed in gym establishments. These processes include membership registration and management, attendance monitoring, payment processing, trainer scheduling, workout plan generation, and customer progress tracking. In many gyms, these operations are still performed manually or through disconnected systems, resulting in delays, duplicate records, scheduling conflicts, and inefficient communication between staff and clients. By integrating these functions into a single intelligent platform, the study seeks to improve workflow efficiency, reduce administrative workload, and provide more accurate and accessible records for both management and clients.

Despite the significant growth in the fitness industry driven by increasing public awareness of health and wellness, many small to medium-sized gym establishments continue to rely on manual or semi-digital systems for managing memberships, payments, attendance, and trainer assignments. These traditional processes result in critical operational inefficiencies, including scheduling conflicts, inaccurate record-keeping, and limited scalability. Furthermore, gym clients increasingly demand customized fitness programs tailored to their specific goals, body composition, health conditions, and experience levels. However, generic routines remain common because manual program design is time-consuming and often leads to inconsistent quality. Additionally, the manual assignment of trainers frequently causes workload imbalances, trainer underutilization, and inconsistent client support, which negatively impacts service quality and overall customer satisfaction.

The partner gym establishment and its clients play a significant role in the development and evaluation of the proposed system. Gym clients, including beginners, intermediate trainees, and experienced fitness enthusiasts, are the primary beneficiaries of the recommendation features of the system. Through the use of collected client data, the system will be capable of generating personalized workout recommendations and improving trainer-client matching. Meanwhile, gym administrators and trainers will benefit from improved scheduling, centralized record management, and automated operational processes that minimize manual tasks and improve decision-making.

To effectively develop, train, and implement this system, specific data will be gathered directly from the partner gym establishment. This includes historical operational records such as membership logs, payment transactions, attendance patterns, and trainer schedules to establish a

baseline for management automation. To fuel the Machine Learning recommendation engine, client-specific data—including body composition metrics, fitness goals, health considerations, and experience levels—will be collected. Finally, trainer profiles comprising availability schedules, areas of specialization, certifications, and past assignment history will be acquired to facilitate scheduling optimization and ensure equitable workload distribution. Through the integration of these datasets, the proposed system aims to deliver a smarter, more efficient, and client-centered gym management solution.

1.2 Statement of the Problem

General Problem

How can a Gym Management System with AI-based features be developed to improve gym operations and provide personalized fitness services for gym clients?

Specific Research Questions

1. How can Artificial Intelligence (AI) be utilized to generate personalized workout recommendations based on user data such as fitness goals, body composition, and experience level?
2. How can the system optimize trainer assignment and scheduling to improve operational efficiency and balanced workload distribution?
3. How can the system provide efficient and centralized access to gym member profiles, attendance records, exercises, meal plans, and other fitness-related information?
4. How can the developed system be evaluated in terms of usability, functionality, and user satisfaction?

1.3 Objectives of the Study

General Objective

To develop a Gym Management System with AI-based features that improves gym operations and delivers personalized fitness services for gym clients.

Specific Objectives

1. Develop a web-based system for managing gym users, memberships, payments, attendance, and trainer scheduling.
2. Integrate an AI-based recommendation feature that generates personalized workout plans based on user data and fitness goals.
3. Provide centralized access to user profiles, exercises, meal plans, attendance records, and other fitness-related information.
4. Implement an optimization feature for trainer assignment and scheduling to ensure efficient workload distribution.
5. Evaluate the developed system in terms of usability, functionality, efficiency, and user satisfaction.

1.4 Scope and Delimitation

A. Scope

The study focuses on the development of a **web-based Gym Management System** designed for administrators, staff, and gym members.

A. Administrator Features

1. **User Authentication & Role Management** – Secure login/logout; role-based access control for Admin and Staff
2. **Member Management (CRUD)** – Add, update, delete, and view member profiles with personal, fitness, and membership details
3. **Membership & Payment Management** – Track plans, monitor payments, and automate expiration alerts
4. **Attendance Monitoring** – Log member check-in and check-out records
5. **Workout Recommendation System (AI-Based)** – Generate personalized workout and meal plans based on member attributes
6. **Trainer Assignment Optimization** – Auto-assign trainers using availability, specialization, and workload constraints
7. **Web-Based Access & Centralized Database** – Browser-based access with real-time data synchronization
8. **Responsive Multi-Device Interface** – Accessible on desktops, laptops, and tablets via local network or internet

B. Staff Features

1. **User Authentication & Role Management** – Secure login/logout with staff-level permissions

2. **Member Management (CRUD)** – View and update member profiles and membership status
3. **Membership & Payment Management** – Track plans, payments, and renewal alerts for assigned members
4. **Attendance Monitoring** – Log and view member check-in/check-out activities
5. **Workout Recommendation System (AI-Based)** – View and assist members with AI-generated workout and meal plans
6. **Web-Based Access & Centralized Database** – Role-limited access to centralized data with real-time updates
7. **Responsive Multi-Device Interface** – Operational access via desktops, laptops, and tablets

C. User (Gym Member) Features

1. **Attendance Participation** – Self-log check-in and check-out via web interface
2. **Workout Recommendation System (AI-Based)** – Receive personalized workout and meal plans based on age, BMI, goals, and experience
3. **Web-Based Access & Centralized Database** – View personal account data with real-time synchronization

4. **Responsive Multi-Device Interface** – Accessible on desktops, laptops, and tablets via browser

The Machine Learning component uses supervised classification algorithms trained on labeled fitness datasets. Input features include age, weight, height, BMI, fitness goal, and experience level.

The output consists of the recommended workout category and intensity level. Model performance is evaluated using accuracy score and cross-validation.

The optimization component applies constraint-based algorithms considering trainer availability schedules, maximum client capacity, and specialization compatibility to generate optimal trainer assignments.

B. Delimitation

The study does not include the following:

1. Mobile Application Version (Web-Based System Only)

The system is developed strictly as a web-based platform accessible through internet browsers. The development of a dedicated mobile application is beyond the scope of this study.

2. Hardware Integration such as RFID or Smart Gym Equipment

The system does not integrate with physical hardware devices such as RFID scanners or biometric systems. Attendance logging is performed manually through the web interface.

3. Integration with External Payment Gateways (e.g., GCash, PayPal)

The system does not support real-time online payment processing through third-party platforms.

Payments are manually recorded within the system.

1.5 Significance of the Study

This study will benefit the following stakeholders:

Future Researchers – The study may serve as a reference for future research involving Machine Learning and optimization algorithms in fitness management systems.

Gym Members – Members receive personalized workout recommendations tailored to their fitness goals and physical attributes.

Gym Owners and Administrators – The system streamlines administrative tasks such as membership tracking, payment monitoring, and attendance management, improving operational efficiency.

Gym Staff – Staff benefit from automated record management and trainer scheduling, reducing human error and scheduling conflicts.

Trainers – The optimization feature ensures fair workload distribution and appropriate trainer assignment based on specialization.

1.6 Definition of Terms

Centralized Database – A unified digital repository wherein all empirical observations, administrative records, and member profiles are consolidated into a singular online structure, facilitating real-time data synchronization across authorized web-based interfaces.

Constraint-Based Algorithms – Mathematical methodologies dedicated to locating an optimal extremum within a feasible solution space that is strictly delineated by rigid boundary conditions, specifically applied in this study to resolve trainer scheduling conflicts and workload distribution.

Cross-Validation – A resampling technique employed to assess model generalizability by partitioning datasets into multiple training and testing subsets, thereby mitigating overfitting risks and providing robust performance estimates for the supervised classification component.

Gym Management and Recommendation System – A composite software architecture designed to streamline administrative workflows and enhance member engagement through the integration of managerial utilities and artificial intelligence-driven advisory capabilities.

Machine Learning – A computational paradigm wherein systems achieve autonomous assimilation of latent structural regularities from empirical corpora, enabling the generation of personalized fitness insights without recourse to explicitly codified procedural logic.

Optimization Algorithms – Heuristic processes employed to extremize objective functions under specific limitations, utilized in this context to automate trainer assignments while satisfying availability and specialization constraints.

Role-Based Access Control (RBAC) – A security architecture that circumscribes computational privileges based upon ontological role designations (Admin, Staff, Member) assigned to users, rather than via direct individual identity mapping, ensuring segregated access to sensitive operational modules.

Supervised Classification – A pedagogical algorithmic framework wherein a model ingests annotated data manifolds to extrapolate predetermined consequents (workout categories and intensity levels) through the minimization of divergent error functions.

Workout Recommendation System – An AI-driven advisory module that extrapolates personalized fitness regimens by analyzing member-specific attributes such as age, biometric data, and stated objectives to prescribe appropriate exercise protocols.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Introduction

This chapter presents the review of related literature and studies relevant to the proposed Gym Management and Recommendation System using Optimization and Machine Learning.

The purpose of this chapter is to provide a comprehensive understanding of existing concepts, technologies, and systems that are closely related to the study.

The review is divided into related literature and related studies, both from local and foreign sources. This helps identify existing solutions, compare methodologies, and determine research gaps. The findings from this chapter serve as the foundation for the development of the proposed system.

2.2 Foreign Literature

Internationally, gym management systems have significantly evolved through the integration of advanced information technologies, enabling organizations to streamline operations and enhance service delivery. In the context of the proposed system titled *A Gym Management and Recommendation System using Optimization and Machine Learning*, these technological advancements serve as the foundation for improving both administrative efficiency and user experience. According to Laudon and Laudon (2021), information systems play a critical role in automating business processes, reducing manual intervention, and

enhancing organizational performance. In gym environments, this translates to efficient handling of membership records, attendance monitoring, payment processing, and performance tracking.

Moreover, Pressman and Maxim (2020) emphasized that web-based applications are essential in modern system development due to their scalability, flexibility, and cross-platform compatibility. These features are highly relevant to the proposed system, as they allow gym administrators and members to access services anytime and anywhere. A web-based gym management system ensures real-time updates, centralized data management, and improved accessibility, which are crucial in delivering a seamless and efficient user experience.

The integration of intelligent systems is further supported by Mitchell (1997), who defined machine learning as a system capable of improving its performance through experience. In the proposed capstone project, machine learning plays a vital role in analyzing user data such as fitness goals, body metrics, and workout history. This enables the system to generate personalized workout plans and recommendations, making fitness programs more adaptive and effective for individual users.

In addition, Han et al. (2012) highlighted the effectiveness of supervised learning in classification and prediction tasks. This approach is particularly applicable to the system, where users can be categorized based on their fitness levels, preferences, and progress. By leveraging supervised learning algorithms, the system can predict suitable workout routines and recommend appropriate training programs, thereby enhancing the accuracy and relevance of its recommendations.

Complementing this, Ricci et al. (2015) discussed the significance of recommendation systems in improving user engagement by providing personalized suggestions. Within the proposed system, recommendation mechanisms can suggest exercises, diet plans, and training schedules tailored to each user's needs. This personalization not only improves user satisfaction but also encourages consistency and motivation in achieving fitness goals.

Furthermore, Hillier and Lieberman (2015) introduced optimization techniques that are essential in solving resource allocation and scheduling problems. In the context of the proposed gym management system, optimization is utilized to efficiently assign trainers, manage class schedules, and allocate gym resources. By applying optimization algorithms, the system ensures balanced workloads, minimizes scheduling conflicts, and maximizes the utilization of available resources.

Lastly, Dix et al. (2004) emphasized the importance of user-centered design in developing interactive systems. A user-friendly interface is a critical component of the proposed system, as it directly affects usability and user satisfaction. By designing an intuitive and accessible interface, the system ensures that both gym staff and members can easily navigate and utilize its features, ultimately improving overall system adoption and effectiveness.

In summary, the proposed *Gym Management and Recommendation System using Optimization and Machine Learning* integrates key concepts from information systems, web development, machine learning, recommendation systems, optimization techniques, and user-centered design. These combined technologies provide a comprehensive solution that enhances operational

efficiency while delivering personalized and data-driven fitness experiences, making the system highly relevant and innovative in the modern fitness industry.

2.3 Review of Related Studies

Foreign Studies

Title of the Study: Intelligent Gym Management System Using IoT and Machine Learning

Zhang et al. (2024) developed an intelligent gym management system integrating Internet of Things (IoT) devices and machine learning algorithms to monitor user activity and automate facility management. The system collected real-time data from wearable sensors and gym equipment, which were processed using a decision tree-based classification model to analyze user performance and recommend exercise routines. The system was tested on 1,200 gym users over a three-month period and achieved an accuracy of 89.6 percent in predicting appropriate workout intensity levels. Additionally, the system reduced manual monitoring tasks by 65 percent and improved user engagement by 38 percent due to personalized recommendations.

Research Gap

The study focuses primarily on IoT-based monitoring and lacks optimization techniques for scheduling trainers and managing gym resources. The proposed system addresses this gap by integrating optimization algorithms for efficient trainer assignment and facility scheduling, alongside machine learning-based recommendations.

Title of the Study: A Web-Based Fitness Management System with Personalized Recommendation Engine

Kim and Park (2023) designed a web-based fitness management platform that incorporates a collaborative filtering recommendation system to suggest workout routines based on user preferences and historical data. The system was evaluated using a dataset of 3,500 users and demonstrated a recommendation accuracy of 87.2 percent based on user satisfaction ratings. The platform also supported membership management, attendance tracking, and billing, resulting in a 52 percent reduction in administrative workload for gym staff.

Research Gap

Although the system provides personalized recommendations, it relies solely on collaborative filtering and does not utilize machine learning models for predictive analysis. The proposed system addresses this limitation by incorporating supervised learning techniques to generate more accurate and adaptive workout plans based on user characteristics.

Title of the Study: Machine Learning-Based Personalized Fitness Recommendation System

Patel et al. (2022) implemented a machine learning-based fitness recommendation system using supervised learning algorithms such as Support Vector Machines (SVM) and Random Forest classifiers. The system analyzed user attributes including age, BMI, fitness goals, and activity levels to generate customized workout plans. Using a dataset of 2,800 participants, the system achieved an F1-score of 90.1 percent in correctly classifying suitable workout programs. The authors also reported increased user adherence to fitness plans by 41 percent compared to non-personalized programs.

Research Gap

The study focuses on recommendation generation but does not include a full gym management system for handling administrative tasks such as scheduling and membership tracking. The proposed system addresses this gap by combining both management functionalities and intelligent recommendation features into a unified platform.

Title of the Study: Optimization Techniques for Sports Facility Scheduling

Torres (2021) applied linear programming and genetic algorithms to optimize scheduling in sports facilities. The system was tested on a dataset involving 15 trainers, 10 classes, and over 500 gym members. Results showed a 34 percent improvement in schedule efficiency and a 27 percent reduction in booking conflicts. The optimization model ensured fair workload distribution among trainers and maximized facility utilization.

Research Gap

While the study effectively applies optimization for scheduling, it does not integrate machine learning for personalized user recommendations. The proposed system addresses this limitation by combining optimization techniques with machine learning to create a more intelligent and comprehensive gym management solution.

Title of the Study: User-Centered Design in Fitness Applications

Brown and Wilson (2020) evaluated the impact of user interface design on fitness application usability using a sample of 600 participants. The study utilized usability testing metrics such as task completion rate, error rate, and user satisfaction scores. Results indicated that applications with intuitive interfaces achieved a 92 percent task completion rate and

significantly higher user satisfaction. The authors emphasized that simplicity, accessibility, and responsiveness are critical factors in system adoption.

Research Gap

The study focuses on user interface design but does not incorporate backend intelligence such as recommendation systems or optimization models. The proposed system addresses this gap by integrating a user-friendly interface with advanced machine learning and optimization functionalities.

Local Studies

Title of the Study: Development of a Web-Based Gym Management System in Metro Manila

Dela Cruz and Santos (2023) developed a web-based gym management system designed to streamline membership registration, attendance tracking, and payment processing for small to medium-sized fitness centers in Metro Manila. The system was implemented in three local gyms and evaluated over a two-month period involving 450 active members. Results showed a 58 percent reduction in administrative workload and a 43 percent improvement in record accuracy compared to manual processes. The system also improved customer satisfaction due to faster transaction processing and easier access to membership information.

Research Gap

The study focuses primarily on administrative automation and does not include intelligent features such as personalized workout recommendations or optimization of trainer schedules.

The proposed system addresses this gap by integrating machine learning and optimization techniques to enhance both user experience and operational efficiency.

Title of the Study: Fitness Tracking and Recommendation Mobile Application Using Rule-Based Approach

Reyes et al. (2022) developed a mobile fitness application that provides workout recommendations using a rule-based system based on predefined conditions such as age, weight, and fitness goals. The system was tested with 300 users and achieved a user satisfaction rate of 85 percent. The application successfully guided users in selecting appropriate exercises; however, recommendations were limited to static rules and lacked adaptability to changing user data.

Research Gap

The study relies on a rule-based approach, which limits the system's ability to learn and adapt over time. The proposed system addresses this limitation by using machine learning algorithms that continuously improve recommendations based on user behavior and data.

Title of the Study: Scheduling System for Fitness Centers Using Optimization Techniques

Garcia and Ramos (2021) developed a scheduling system for fitness centers using basic optimization techniques to assign trainers and allocate workout sessions. The system was tested in a local gym with 12 trainers and 200 members, resulting in a 29 percent reduction in scheduling conflicts and improved trainer utilization. The authors emphasized the importance of systematic scheduling in improving service efficiency.

Research Gap

While the study applies optimization in scheduling, it does not integrate personalized recommendation systems for users. The proposed system addresses this gap by combining optimization for resource management with machine learning for personalized fitness recommendations

Title of the Study: Design and Implementation of a Human Resource Management System with Recruitment Module

Torres et al. (2024) developed a web-based human resource management system that includes employee scheduling, attendance tracking, and performance monitoring. The system was evaluated in a local organization with 150 employees and resulted in a 47 percent improvement in task efficiency. The study highlighted the importance of centralized systems in improving organizational workflows and decision-making processes.

Research Gap

The study focuses on employee management and does not address fitness-specific functionalities such as workout tracking or recommendation systems. The proposed system addresses this limitation by applying similar management principles within a gym context while integrating intelligent recommendation and optimization features.

Title of the Study: Adoption of Information Systems in Small and Medium Enterprises in the Philippines

Lopez and Mercado (2020) conducted a survey involving 210 small and medium enterprises (SMEs) in the Philippines to assess the adoption of information systems. The study

found that 76 percent of SMEs reported improved operational efficiency after implementing digital systems. However, only 28 percent utilized advanced technologies such as data analytics or machine learning. The authors concluded that while digital adoption is increasing, the use of intelligent systems remains limited.

Research Gap

The study highlights limited adoption of advanced technologies but does not provide a specific system implementation that integrates these technologies. The proposed system addresses this gap by incorporating machine learning and optimization into a practical gym management platform.

Title of the Study: Usability Evaluation of Mobile Health and Fitness Applications in the Philippines

Castro et al. (2023) evaluated the usability of mobile fitness applications among 500 Filipino users using the System Usability Scale (SUS). The study found that applications with simple navigation and personalized features achieved higher usability scores, averaging 88 out of 100. Users expressed a strong preference for applications that adapt to their fitness goals and provide tailored recommendations.

Research Gap

Although the study emphasizes usability and personalization, it does not present a system that combines these features with backend optimization and intelligent decision-making. The proposed system addresses this by integrating user-centered design with machine learning and optimization techniques.

2.4 Synthesis of the Review of Related Literature and Studies

The reviewed foreign and local studies collectively demonstrate the growing importance of integrating information systems, machine learning, and optimization techniques in developing efficient and intelligent management systems. Foreign studies emphasized the effectiveness of machine learning algorithms in generating personalized recommendations, particularly in fitness applications where user-specific data such as physical attributes, goals, and activity history are utilized to improve accuracy and engagement. Additionally, studies on optimization techniques highlighted their significant role in solving scheduling and resource allocation problems, leading to improved operational efficiency and reduced conflicts in system management.

Moreover, several foreign studies focused on the development of web-based gym and fitness systems, highlighting their scalability, accessibility, and ability to centralize data management. These systems were shown to significantly reduce administrative workload while enhancing user experience through automation and real-time access. However, most of these systems either focused solely on recommendation mechanisms or on administrative functionalities, lacking a fully integrated approach that combines both intelligent recommendations and optimized resource management.

On the other hand, local studies revealed that many gym management and fitness-related systems in the Philippines primarily focus on basic functionalities such as membership management, attendance tracking, and scheduling. While these systems have successfully improved operational efficiency and reduced manual processes, they generally lack advanced

features such as machine learning-based personalization and optimization-driven decision-making. Additionally, some local studies utilized rule-based approaches for recommendations, which are limited in adaptability and do not improve over time compared to machine learning models.

Furthermore, both foreign and local studies emphasized the importance of user-centered design in ensuring system usability and user satisfaction. Systems with intuitive interfaces and personalized features were found to have higher adoption rates and better user engagement.

However, there remains a gap in combining usability with advanced backend intelligence, such as integrating recommendation systems with optimization algorithms in a single platform.

Based on these findings, a significant research gap exists in the development of a comprehensive gym management system that integrates administrative functions, machine learning-based recommendation systems, and optimization techniques within a unified platform. Most existing systems address only specific components of gym operations, resulting in fragmented solutions that limit overall efficiency and user experience.

Therefore, the proposed study, *A Gym Management and Recommendation System using Optimization and Machine Learning*, aims to address these gaps by developing an integrated system that combines automated gym management functionalities, intelligent workout recommendation through machine learning, and efficient resource allocation using optimization techniques. This approach ensures a more holistic, adaptive, and efficient solution that enhances both operational performance and user satisfaction in modern fitness environments.

CHAPTER III

RESEARCH METHODOLOGY

3.1 Research Method

This study will utilize literature review and analysis of existing gym management practices as the primary research approach to gather essential information for the proposed Gym Management and Recommendation System using Optimization and Machine Learning. The research will begin by identifying the problem through examining the challenges encountered in current gym management systems, such as manual membership tracking, inefficient equipment maintenance scheduling, lack of personalized workout recommendations, and suboptimal resource allocation. These issues will serve as the basis for proposing a more efficient, data-driven, and intelligent gym management solution. A comprehensive review of related literature, existing gym management systems, and machine learning-based recommendation technologies will be conducted to identify the strengths and weaknesses of current systems, particularly the limitation of most solutions focusing only on administrative tasks without integrating intelligent optimization and personalized recommendation features. In addition, local studies will be analyzed to understand the current state of technology adoption in fitness centers, where many organizations still lack integrated systems that combine operational management with AI-powered personalization. Through the analysis and synthesis of information from scholarly articles, journals, and case studies, the researchers will establish a strong foundation for the proposed system following a systematic development cycle: planning by identifying system objectives and user requirements, designing the system architecture and machine

learning models, developing the system modules including user management and AI-based recommendations, testing functionality and optimization accuracy, deploying the system in a real gym environment, and reviewing performance for continuous improvement. This research method will ensure that the proposed system is relevant, efficient, and aligned with real-world gym operational needs while addressing the limitations identified in existing systems and related studies through the integration of optimization algorithms and machine learning techniques for enhanced member experience and operational efficiency.



Figure 1: Agile Methodology

The methodology adopted in this study is the Agile methodology, chosen for its iterative, adaptive, and collaborative nature in software development. Given the complexity of developing an Automated Candidate Screening System with integrated Resume Analysis and Video Interview Evaluation using Natural Language Processing (NLP), Agile supports the systematic

construction of each system component through short, manageable development cycles called sprints. Each sprint produces a testable version of a system module such as the Resume Analysis Module, the Video Interview Evaluation Module, or the HR Dashboard, enabling continuous refinement based on stakeholder feedback. The Agile approach is particularly suited to this study because the system's requirements involve multiple interconnected features that benefit from iterative testing and incremental improvement rather than a single rigid development cycle.

Context Diagram

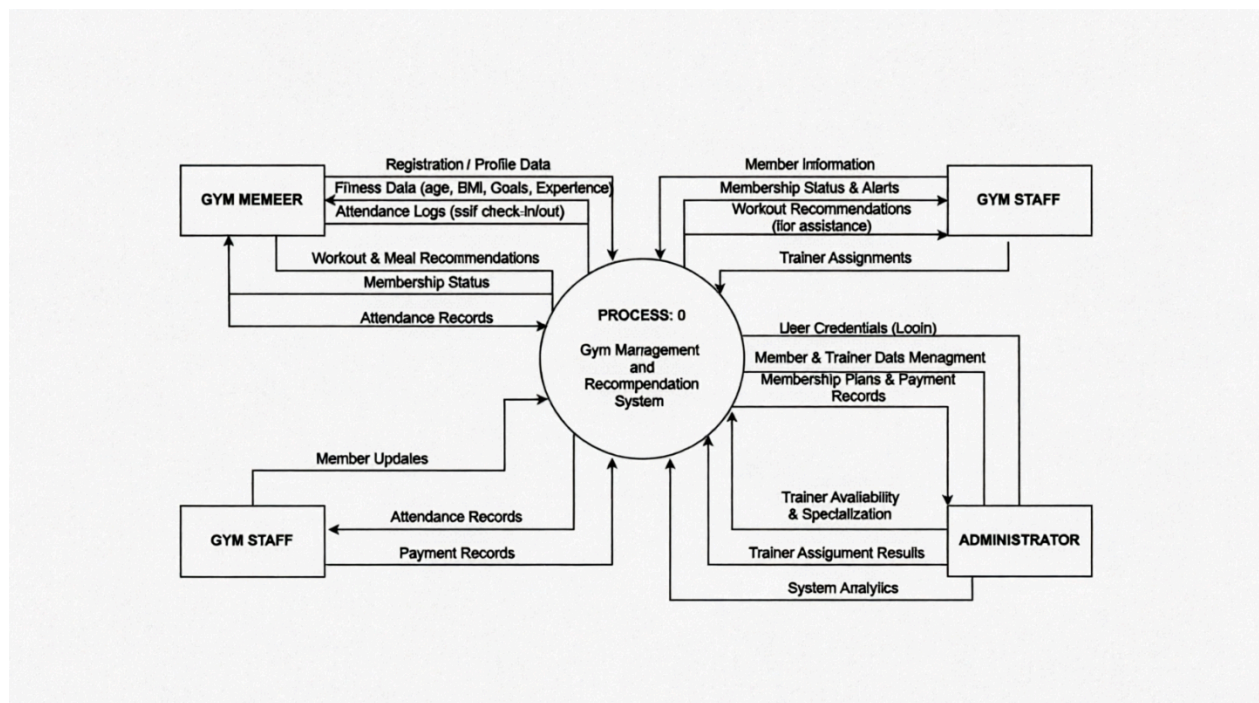


Figure 2: DFD Level 0 (Context Diagram) Proposed System

Figure [X] presents the Context Diagram (Level 0 Data Flow Diagram) of the proposed Gym Management and Recommendation System. The diagram illustrates the system as a single process interacting with external entities, namely the Gym Member, Gym Staff, and Administrator. The Gym Member provides input data such as registration and profile information, fitness metrics including age, BMI, goals, and experience, along with

self-check-in/out attendance logs, while receiving outputs including workout and meal recommendations, membership status, and attendance records. The Gym Staff supplies member updates and receives detailed information such as member profiles, membership status alerts, workout recommendations for client assistance, trainer assignments, attendance logs, and payment records. Meanwhile, the Administrator provides user credentials, manages member and trainer data, configures membership plans and payment records, defines trainer availability and specialization, and receives trainer assignment results and system analytics. This diagram provides a high-level overview of data exchange between the system and its external users.

Design

In this phase, the collected requirements will be translated into a structured system design. The researchers will develop the system architecture, database schema, UML diagrams, and interface prototypes. This phase will ensure that all system components, including the User Management Module, Membership Module, Attendance Tracking Module, Trainer Assignment Module, and AI-based Recommendation Module, are properly planned before implementation.

Data Flow Diagram

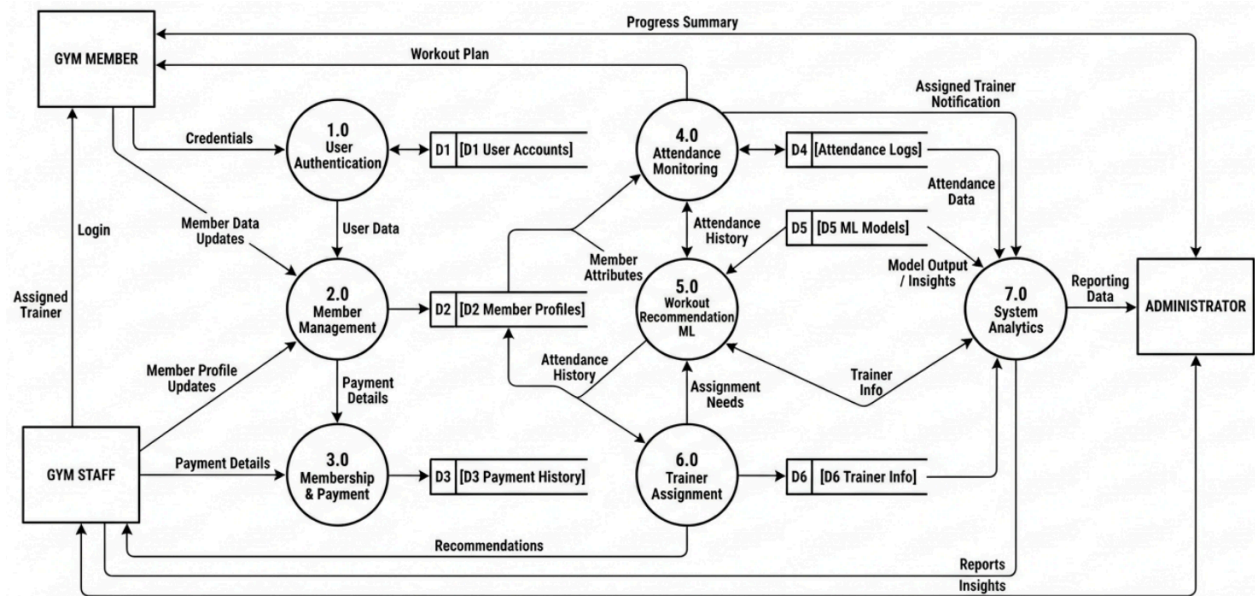


Figure 3: DFD Level 1 Proposed System

Figure 3 depicts the Level 1 Data Flow Diagram for the Gym Management and Recommendation System. It details the primary operational workflows, including identity verification, member and financial administration, trainer scheduling, attendance tracking, AI-powered workout generation, and system reporting. The illustration maps the movement of data between external users—specifically Gym Members, Staff, and Administrators—and the system's internal mechanisms. It highlights how the platform captures inputs like credentials and usage logs to generate outputs such as personalized fitness plans and management insights. This breakdown demonstrates how the system separates complex operations into distinct, manageable functions to support both efficient facility management and tailored user experiences.

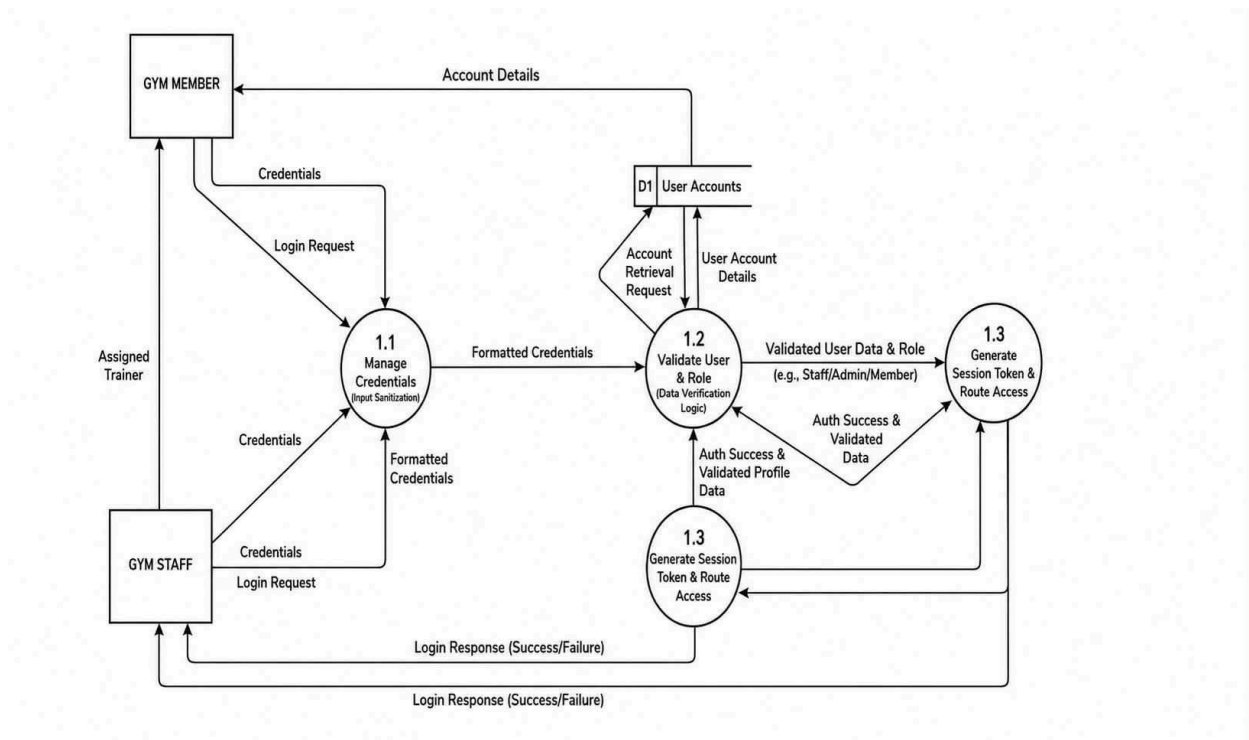


Figure 4: DFD 1.0 Level 2 User Authentication

Figure 4 illustrates the Level 2 Data Flow Diagram of Process 1.0, detailing the User Authentication subsystem of the Gym Management and Recommendation System. It presents the core authentication workflows, including credential management and input sanitization, user validation and role verification, and session token generation with route access control. The diagram shows how data flows between external entities such as Gym Members and Gym Staff, as well as how login requests are processed through multiple validation stages before granting system access. This level provides a granular view of how the authentication process is decomposed into sequential sub-processes to ensure secure user verification, proper role assignment, and protected access to system features based on user credentials and permissions.

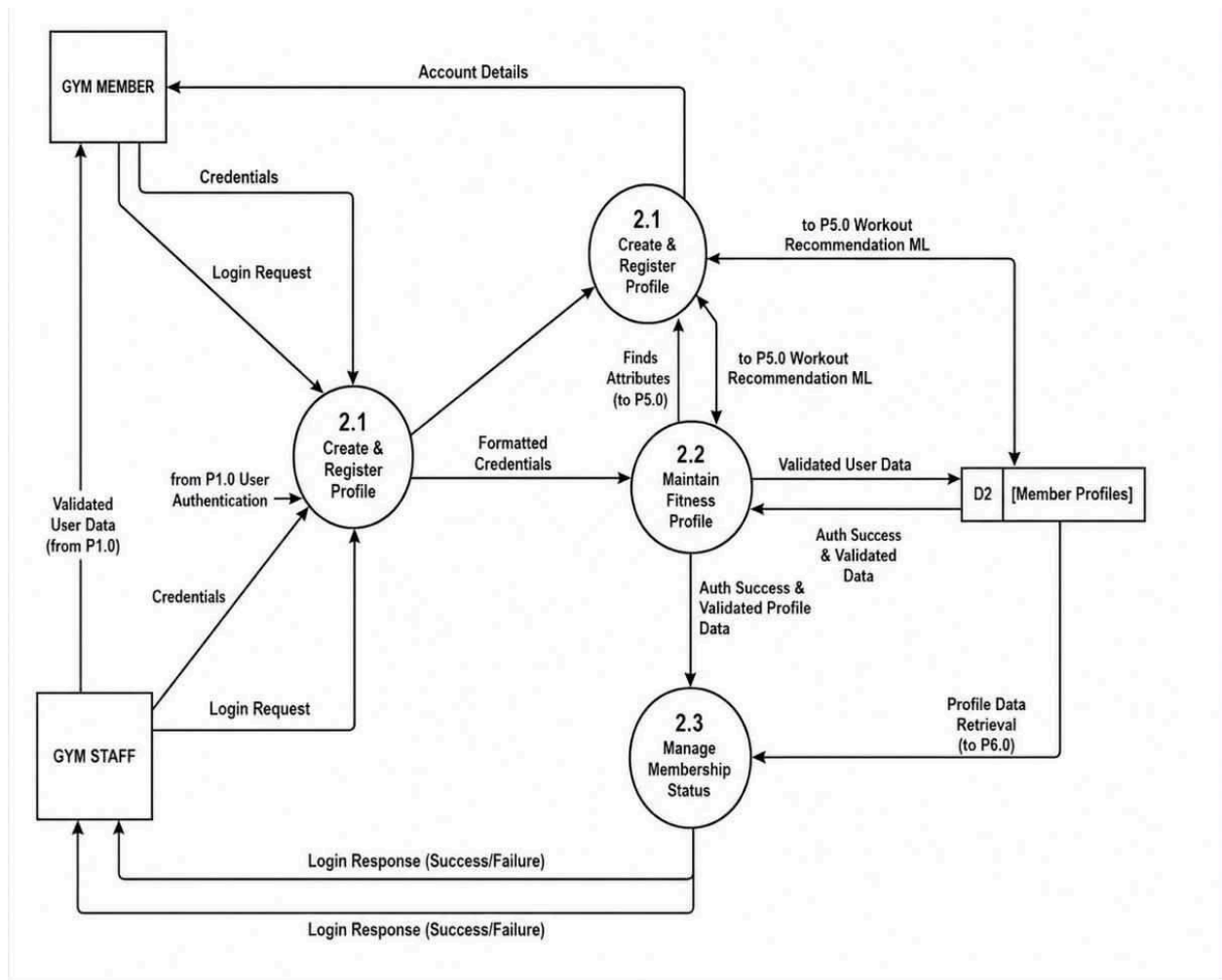


Figure 5: DFD 2.0 Level 2 Member Management

Figure 5 illustrates the Level 2 Data Flow Diagram of Process 2.0, the Member Management subsystem of the proposed Gym Management and Recommendation System. It presents the major sub-processes involved in handling member data, including profile creation and registration, fitness attribute maintenance, and membership status management. The diagram shows how data flows between external entities such as Gym Members and Gym Staff, as well

as how user information is stored in the Member Profiles database and distributed to other functional modules like the Workout Recommendation ML and System Analytics. This level provides a detailed overview of how the member management process is decomposed into specific functional components to ensure accurate profile tracking, data integrity, and the effective utilization of member data for personalized services and administrative reporting.

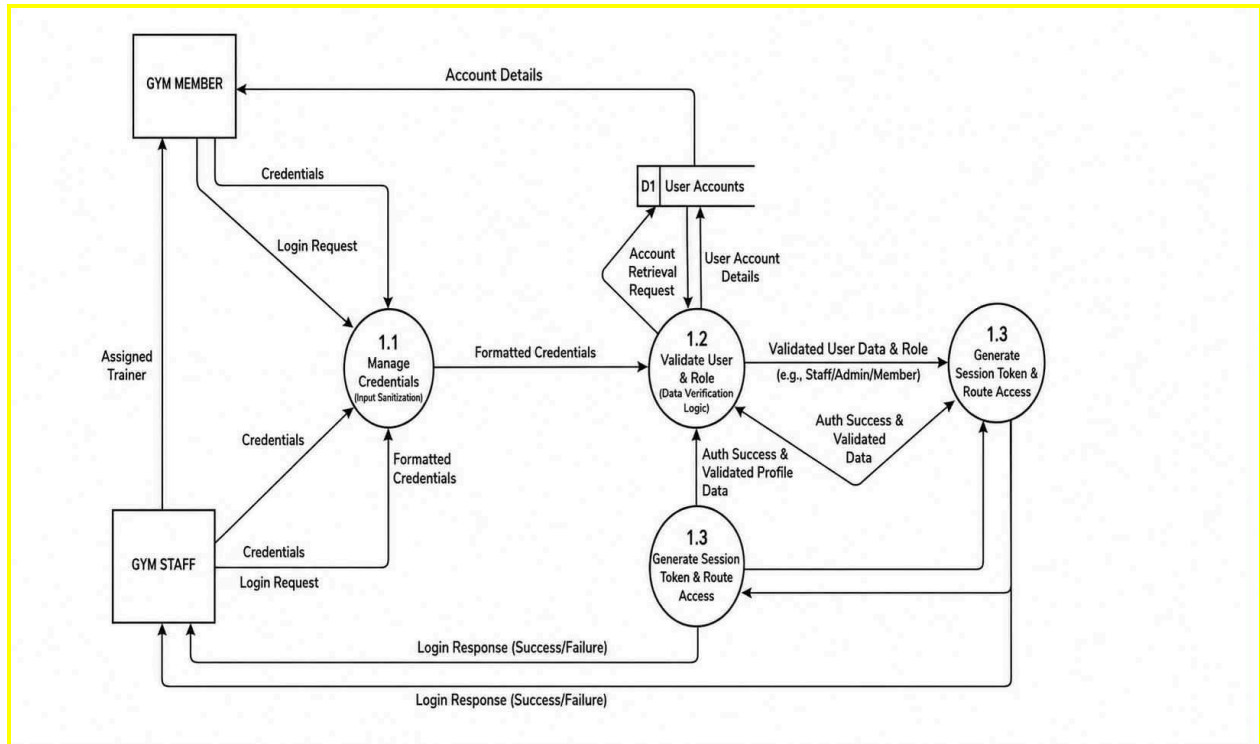


Figure 6: DFD 3.0 Level 2 Membership and Payment

Figure 6 illustrates the Level 2 Data Flow Diagram of Process 3.0, the Membership and Payment subsystem of the proposed Gym Management and Recommendation System. It presents the major processes involved in the system, including plan selection and modification, transaction detail handling, and the updating of payment records. The diagram shows how data flows between external entities such as Gym Members, Gym Staff, and the Administrator, as well as how the system processes plan requests and financial data against stored plan definitions. This level provides a detailed overview of how the payment and membership process is decomposed

into functional components to ensure accurate billing, efficient plan management, and the generation of consolidated financial data for administrative oversight and system analytics.

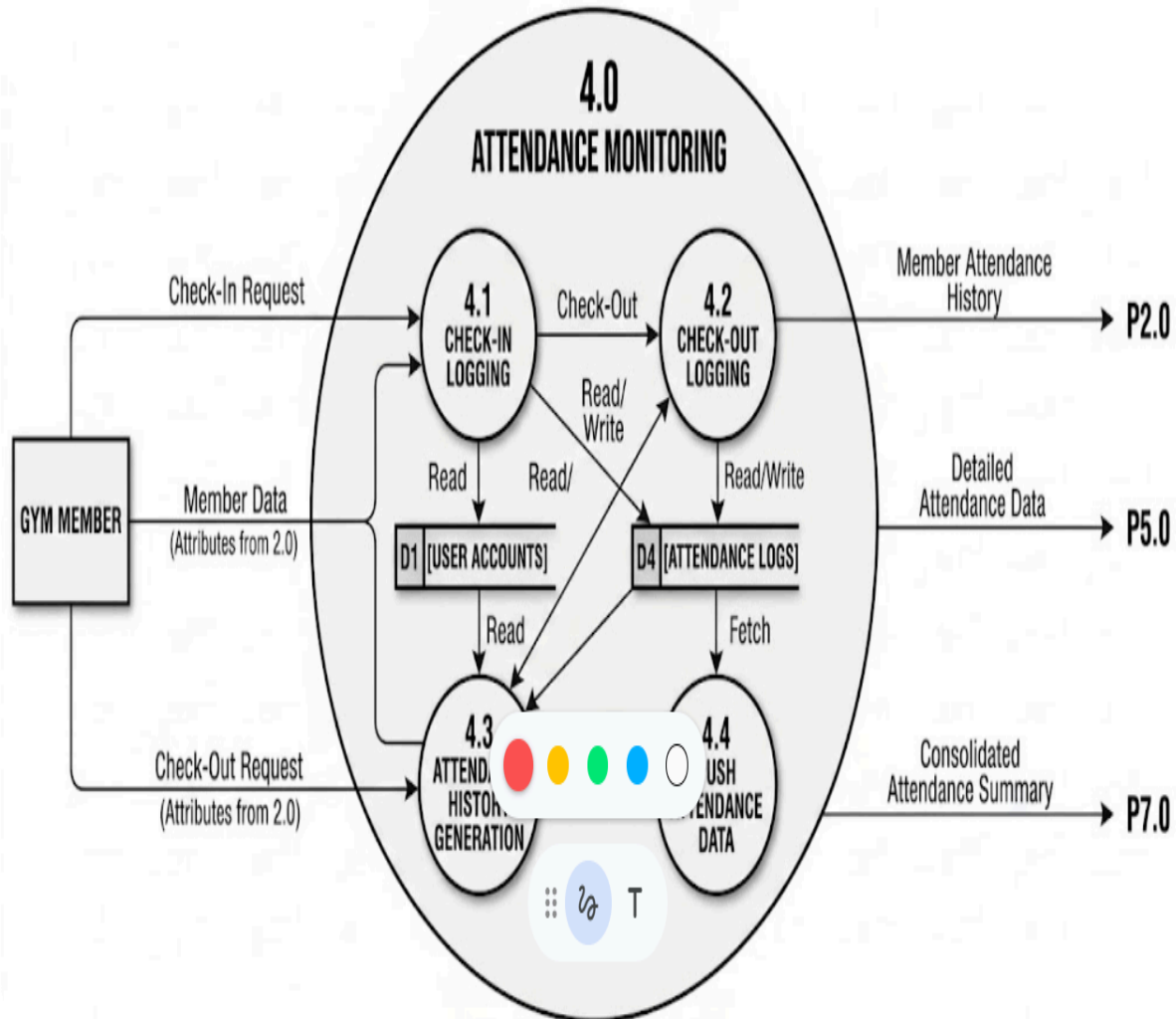


Figure 7: DFD 4.0 Level 2 Attendance and Monitoring

Figure 7 illustrates the Level 2 Data Flow Diagram of Process 4.0, the Attendance Monitoring subsystem of the proposed Gym Management and Recommendation System. It presents the major sub-processes involved in tracking user activity, including check-in and check-out logging, the generation of attendance history, and the automated distribution of attendance data.

The diagram shows how data flows from the Gym Member into the system to be validated against D1 [User Accounts] and recorded within D4 [Attendance Logs]. This level provides a detailed overview of how the monitoring process is decomposed into functional components to ensure precise tracking of gym usage. Furthermore, it highlights how consolidated attendance data is pushed to Process 5.0 (Workout Recommendation ML) for model refinement and Process 7.0 (System Analytics) for administrative reporting, ensuring that member consistency is integrated into the system's intelligent recommendations and operational oversight.

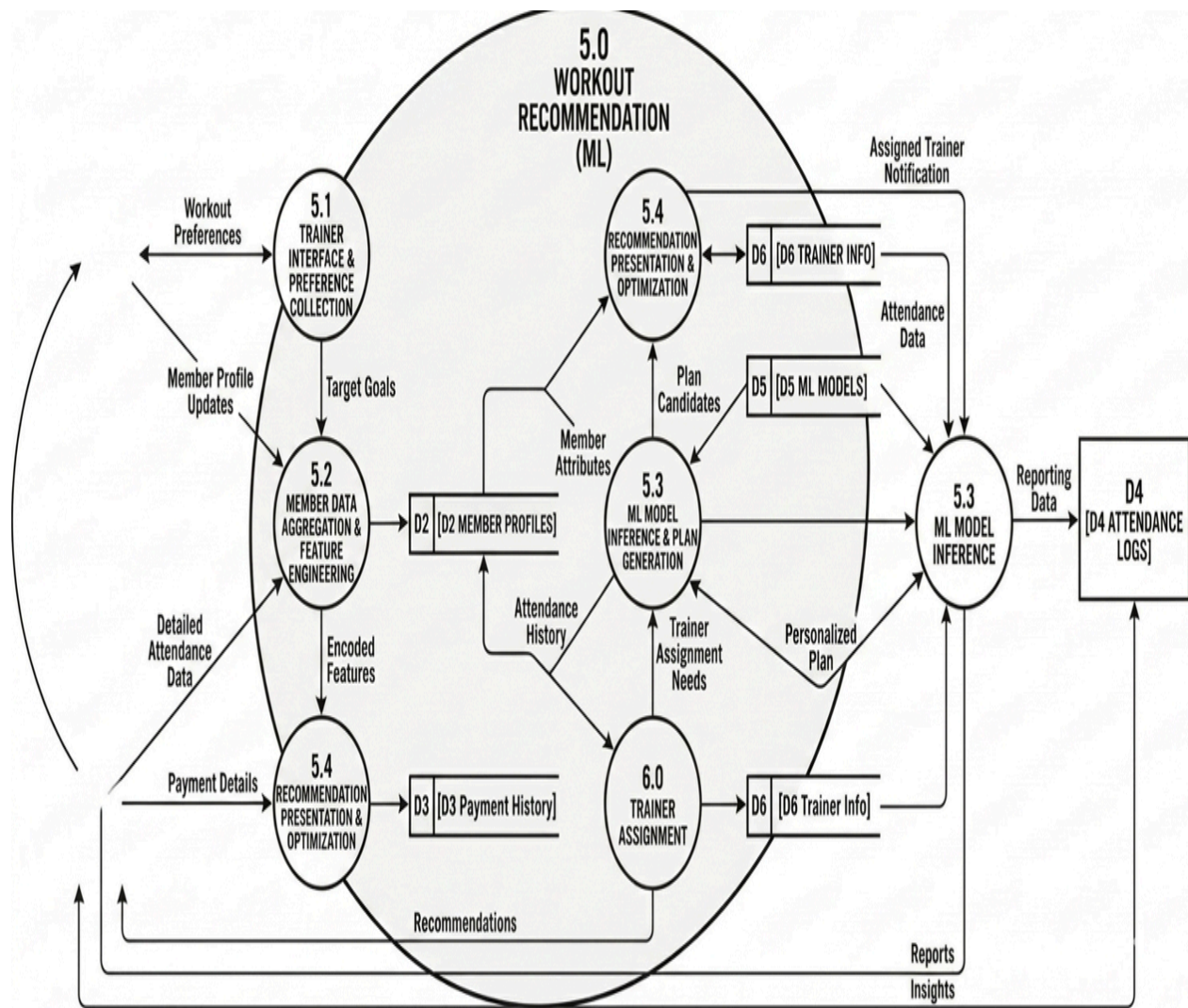


Figure 8: DFD 5.0 Level 2 Workout Recommendation

Figure 8 illustrates the Level 2 Data Flow Diagram of Process 5.0, the Workout Recommendation (ML) subsystem of the proposed Gym Management and Recommendation System. It presents the major sub-processes involved in the intelligent generation of workout plans, including feature engineering, model inference, and recommendation optimization.

The diagram shows how data flows from D2 [Member Profiles] and D4 [Attendance Logs] to be processed by the ML Model Inference engine to generate personalized fitness strategies. This level provides a detailed overview of how the recommendation process is decomposed into functional components to ensure that user-specific attributes and historical performance data are accurately translated into optimized workout plans. Additionally, it demonstrates the integration of D5 [ML Models] for predictive accuracy and the feedback loop to D6 [Trainer Info], ensuring that the generated plans are aligned with available professional guidance and system analytics for continuous improvement.

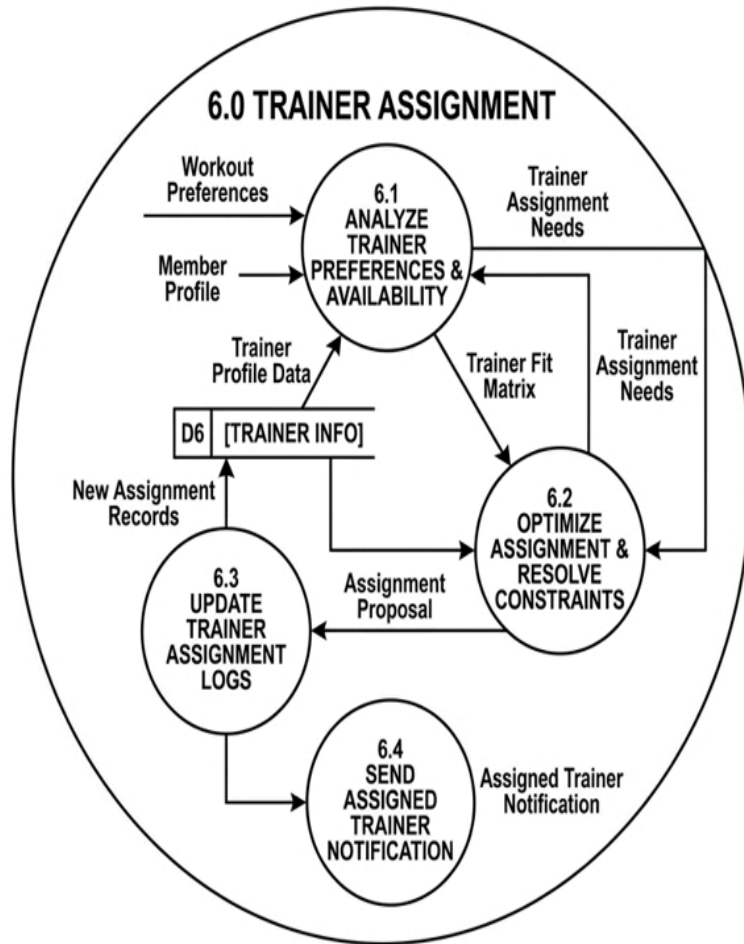


Figure 9: DFD 5.0 Level 2 Trainer Assignment

Figure 9 illustrates the Level 2 Data Flow Diagram of Process 6.0, the Trainer Assignment subsystem of the proposed Gym Management and Recommendation System. It presents the major sub-processes involved in matching members with appropriate coaches, including trainer availability analysis, assignment optimization, and the updating of coach-member records.

The diagram shows how data flows between the Member Profile and the stored trainer data in D6 [Trainer Info] to resolve scheduling constraints and match specific fitness goals with coach specializations. This level provides a detailed overview of how the assignment process is

decomposed into functional components to ensure an efficient and optimized allocation of resources. Furthermore, it demonstrates how the system finalizes the assignment by updating the relevant data stores and triggering notifications to both the trainer and the gym member, ensuring a seamless integration of human-led coaching within the automated recommendation framework.

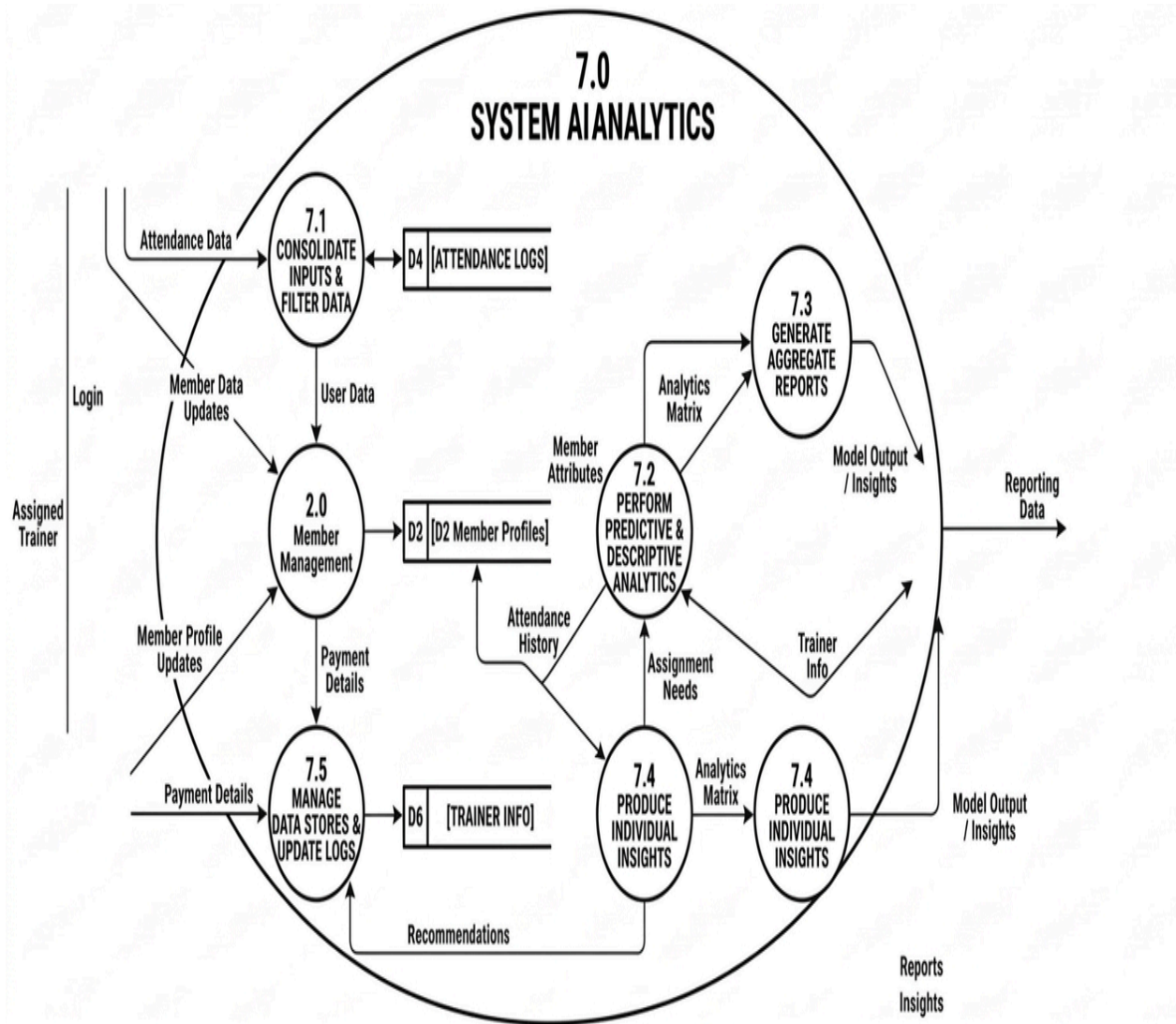


Figure 10: DFD 5.0 Level 2 System Analytics

Figure 10 illustrates the Level 2 Data Flow Diagram of Process 7.0, the System Analytics subsystem of the proposed Gym Management and Recommendation System. It presents the major sub-processes involved in data synthesis and administrative reporting, including the consolidation of attendance logs, the execution of predictive and descriptive analytics, and the generation of both aggregate and individual insights.

The diagram shows how data flows from various stores, such as D4 [Attendance Logs] and D2 [Member Profiles], to be processed into actionable reporting data for the Administrator. This level provides a detailed overview of how the analytics process is decomposed into functional components to ensure that raw system data is transformed into meaningful metrics regarding gym operations and member progress. Additionally, it highlights the system's ability to produce consolidated reports and specialized insights, ensuring comprehensive oversight and data-driven decision-making for the management of the facility.

Development During this phase, the system analytics module will be implemented using iterative Agile sprints. The researchers will develop the core components of the analytics engine, including automated attendance data consolidation, predictive member progress modeling, and administrative dashboard functionalities. Continuous integration and refinement of the machine learning algorithms will be performed throughout development to ensure that the generated insights remain aligned with the gym's operational requirements and member fitness goals.

Testing

The analytics module will undergo rigorous testing to ensure that all data synthesis processes function correctly and meet the defined requirements. Functional testing, integration testing, and system testing will be conducted on all features, including the processing of raw attendance logs, the accuracy of predictive analytics, and the generation of consolidated reporting data. Feedback from gym administrators and trainers will be gathered to validate that the system insights align with actual fitness trends and gym workflows, and necessary improvements will be applied to the data processing logic accordingly.

Deployment

The analytics subsystem will be deployed in a local staging environment to simulate real-world usage conditions. Gym staff and selected administrators will interact with the reporting dashboard to evaluate its performance, usability, and data accuracy in an operational setting. The feedback gathered regarding the clarity of the visual reports and the relevance of the member insights will be used to implement final refinements to the reporting logic prior to full system deployment.

Review

The system analytics module will be evaluated after deployment to assess its overall effectiveness in providing data-driven oversight. User feedback from gym administrators will be analyzed to determine the system's accuracy in predicting peak hours, its usability in tracking member consistency, and its efficiency in improving the gym's resource management. The

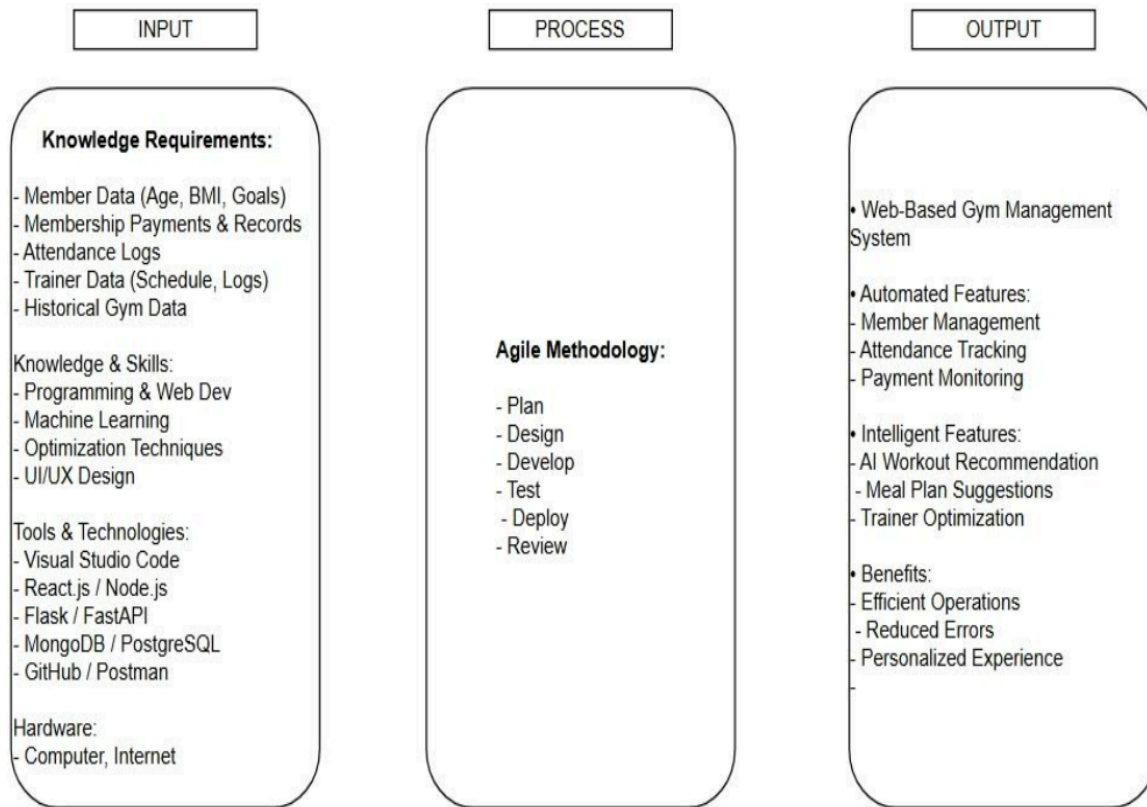
findings from this phase will be used to identify potential enhancements for the optimization algorithms and provide recommendations for future system improvements.

3.2 Conceptual Framework of the Study

Figure 11 illustrates the conceptual framework of the AI-Powered Gym Management and Recommendation System. The framework serves as a guide in the development of the proposed system designed to streamline fitness tracking and personalized workout programming. The platform integrates structured member profiling, automated attendance monitoring, and machine learning-driven workout recommendations to assist gym facilities in managing their members more efficiently.

Through the use of Artificial Intelligence, the system analyzes member attributes and historical performance data, generates optimized fitness plans, and assigns appropriate trainers based on availability and specialization. This enables gym administrators and coaches to manage member progress and facility operations more quickly, consistently, and effectively, ensuring a data-driven approach to health and physical conditioning.

Input Process Output (IPO)



The Figure 12 Conceptual Framework illustrates the development of the proposed system using the Input–Process–Output (IPO) model. This framework serves as a guide in identifying the system requirements, development process, and expected output.

The **Input** stage includes the knowledge requirements, technical skills, and the hardware and software tools necessary for system development. The knowledge requirements consist of member data (age, BMI, fitness goals), membership payments and records, attendance logs, and trainer-specific data such as schedules. These are supported by the researchers' skills in programming, machine learning, and UI/UX design. The proponents utilize technologies such as React.js/Node.js for the frontend and backend, Flask or Fast API for the machine learning

integration, and PostgreSQL or MongoDB for database management to implement the proposed system.

The **Process** stage involves the use of the Agile methodology, which follows iterative phases including planning, design, development, testing, deployment, and review. This iterative cycle ensures that the system's core features, particularly the recommendation engine and automated tracking modules, are continuously refined and aligned with the actual operational needs of the gym environment.

The **Output** stage is the developed Web-Based Gym Management and Recommendation System. The system features automated management functions such as attendance tracking and payment monitoring, alongside intelligent features like AI-driven workout recommendations, meal plan suggestions, and trainer optimization. The system aims to provide efficient gym operations, reduced manual errors, and a personalized fitness experience for all gym members.

3.3 Data Gathering Tools and Procedure

The development of the Web-Based Gym Management and Recommendation System required a comprehensive understanding of current fitness facility operations to ensure that the proposed system is relevant, efficient, and aligned with real-world needs. To achieve this, the researchers utilized a combination of data gathering methods:

- ***Interviews***

Interviews were conducted with gym managers and administrative staff to obtain firsthand information regarding the current gym workflow. This method was used to understand how manual membership tracking and payment recording are performed, identify inefficiencies in the

current process, and determine the specific administrative requirements that the proposed system must address.

- ***Focus Group Discussions***

Focus group discussions were carried out with gym trainers and active members to gather multiple perspectives on workout programming and coach assignments. This method was used to identify common issues, such as the lack of personalized fitness plans and scheduling conflicts, to ensure the system's recommendation and optimization features are tailored to user expectations.

- ***Observation***

Direct observation was conducted within the gym premises to examine the actual daily operations in practice. This method was used to verify real-world workflows, such as the manual logging of attendance during peak hours, ensuring that the proposed system design is based on actual operational conditions and can effectively handle high volumes of data.

Context Diagram

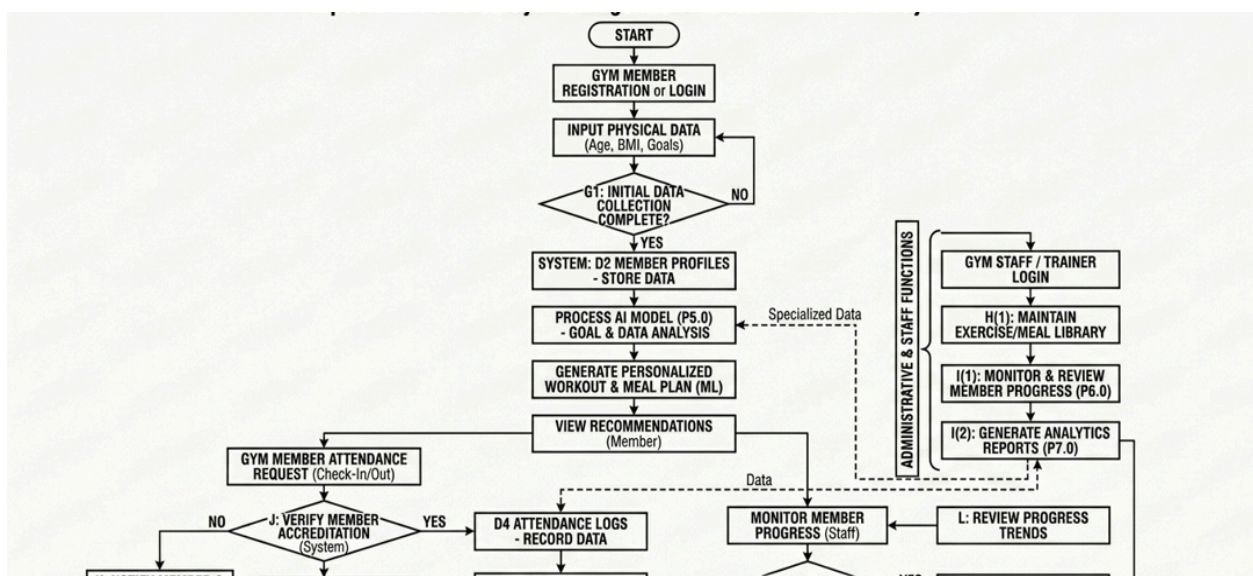


Figure 13: Process Flowchart of the Existing System

Figure 13 illustrates the Process Flowchart of the proposed Gym Management and Recommendation System. The framework serves as a guide for the system's operational logic, detailing how user interactions, data storage, and machine learning models work together to automate gym management.

The process begins with Gym Member Registration or Login, where users provide essential physical data such as age, BMI, and fitness goals. If the initial data collection is complete, the system proceeds to store this information in D2 [Member Profiles]. This data is then fed into the AI Model (Process 5.0) for goal and data analysis, which triggers the generation of personalized workout and meal plans. Once processed, the member can view these intelligent recommendations through the system interface.

Simultaneously, the system handles operational tasks such as Attendance Requests. When a member attempts to check in or out, the system verifies their accreditation. Upon successful verification, the data is recorded in D4 [Attendance Logs], allowing members to track their history while providing the system with performance data for Continuous Plan Optimization. If verification fails, the system notifies the member or allows for manual staff intervention to ensure accurate record-keeping.

On the administrative side, Gym Staff and Trainers can log in to maintain the exercise and meal libraries, monitor member progress, and generate analytics reports through Process 7.0. These reports provide insights into progress trends, allowing staff to assist in inputting updated data when necessary. This cycle ensures that the system remains dynamic, constantly refining its

recommendations based on real-world usage and professional oversight to provide an efficient and personalized gym experience.

3.4 Project Design

System Architecture Diagram

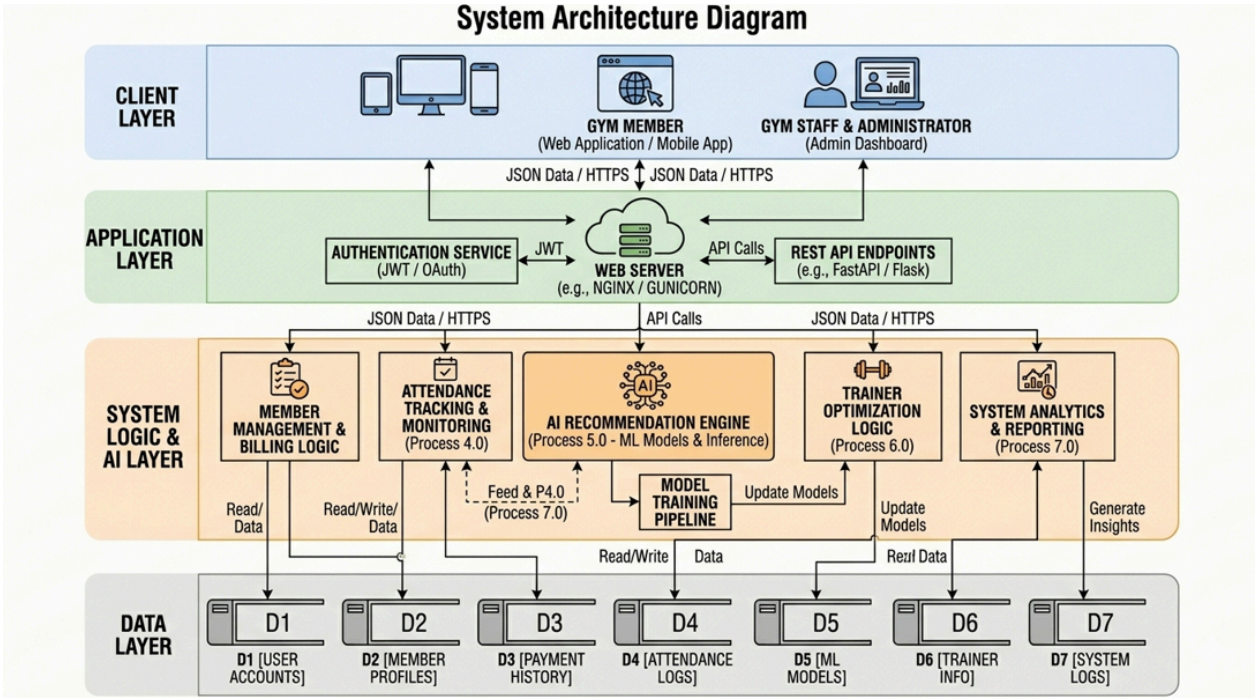


Figure 14: System Architecture Design

Figure 14 illustrates the System Architecture of the proposed Gym Management and Recommendation System. The architecture is organized into four distinct layers to ensure a modular, scalable, and secure environment for processing both administrative tasks and intelligent fitness recommendations.

The Client Layer serves as the primary interface for users, including Gym Members who access the web application or mobile app and the Gym Staff and Administrators who use the specialized Admin Dashboard. These users interact with the system by sending and receiving data through HTTPS protocols, typically in JSON format, to ensure smooth communication across different devices.

The Application Layer acts as the gateway between the users and the core system logic. It includes an Authentication Service that utilizes secure tokens like JWT to manage user sessions and permissions. A Web Server manages the traffic and routes API calls to the appropriate REST API Endpoints, developed using frameworks such as FastAPI or Flask. This layer ensures that all requests are validated and efficiently distributed to the underlying services.

The System Logic and AI Layer contains the core functional modules of the application. This includes the Member Management and Billing Logic, Attendance Tracking, and the central AI Recommendation Engine. This layer is responsible for executing the Machine Learning models that analyze member attributes to generate fitness plans. It also features a Model Training Pipeline and System Analytics logic to ensure that recommendations are constantly refined and administrative insights are generated based on real-world usage patterns.

Finally, the Data Layer provides the foundational storage for all system information. It is composed of multiple data stores including D1 for User Accounts, D2 for Member Profiles, D3 for Payment History, and D4 for Attendance Logs. It also houses the ML Models in D5 and

Trainer Information in D6. This structured data storage allows the system logic to perform efficient read and write operations, ensuring data integrity and consistency across all management and recommendation functions.

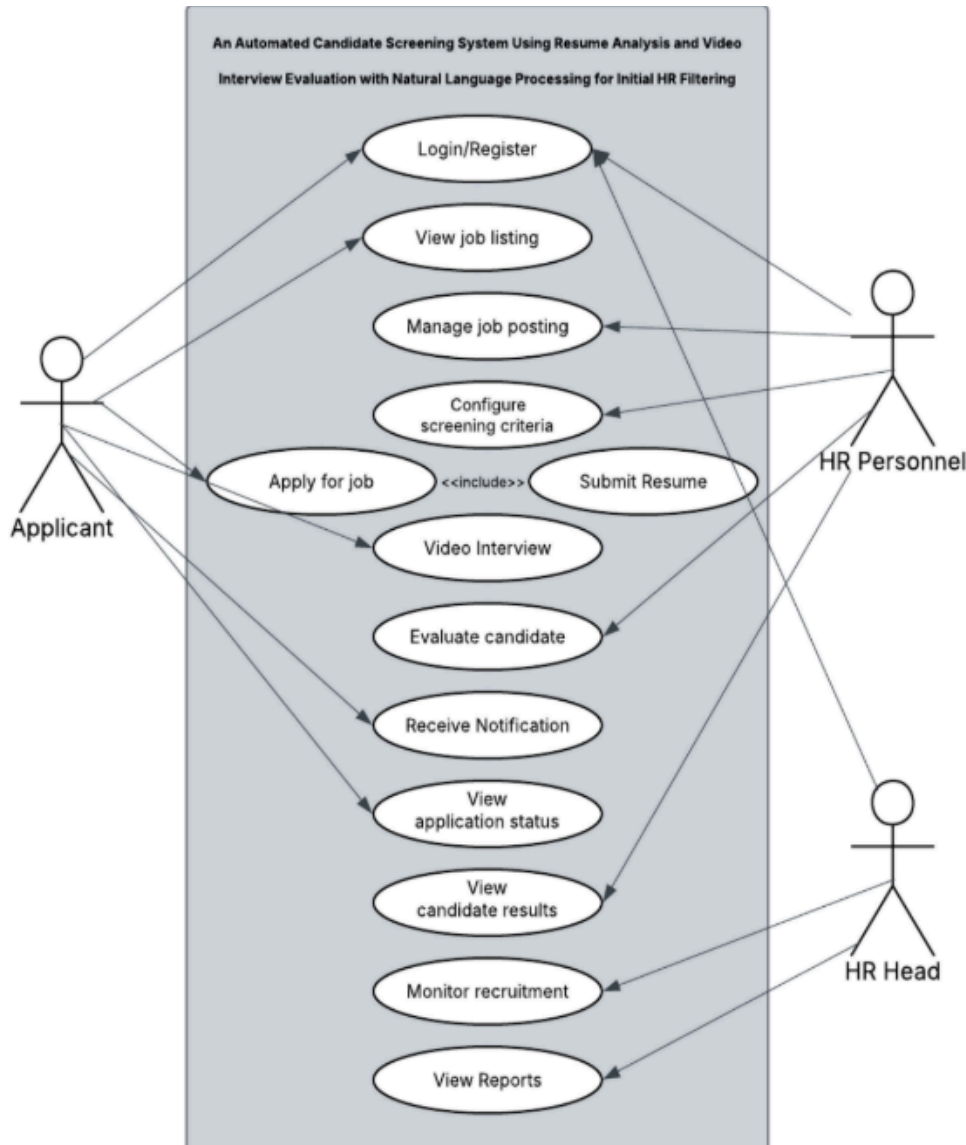


Figure 14: Proposed Use Case

Figure 15 illustrates the Use Case Diagram of the proposed Web-Based Gym Management and Recommendation System. This diagram identifies the primary actors of the system and their specific interactions with the system's functional modules.

The Gym Member is the primary user actor who interacts with the system to manage their fitness journey. Their use cases include logging into the system, viewing their profile, and logging workout and meal data. Most importantly, the member receives personalized workout

and meal recommendations generated by the AI, and they can view the gym schedule and receive notifications regarding their membership or training sessions.

The Gym Staff or Trainer actor focuses on the operational and instructional side of the facility. They are responsible for monitoring progress trends of the members, updating trainer assignments and schedules, and recording attendance. They also have the authority to view and log data alongside the member to ensure that the training information is accurate and professionally validated.

The ML/AI Model is represented as an external system actor that interacts with the core recommendation use cases. It processes the raw data provided by members and staff to generate the specialized recommendations, ensuring that the intelligent aspect of the system remains dynamic and data-driven.

The Administrator actor manages the high-level system functions and facility oversight. Their use cases involve generating system analytics and reports to monitor gym performance. They also configure the recommendation parameters to ensure the AI logic remains accurate and manage the general gym facilities and equipment data. Through the include relationship, the administrator also manages system settings to ensure the overall security and stability of the platform.

Entity Relationship Diagram (ERD)

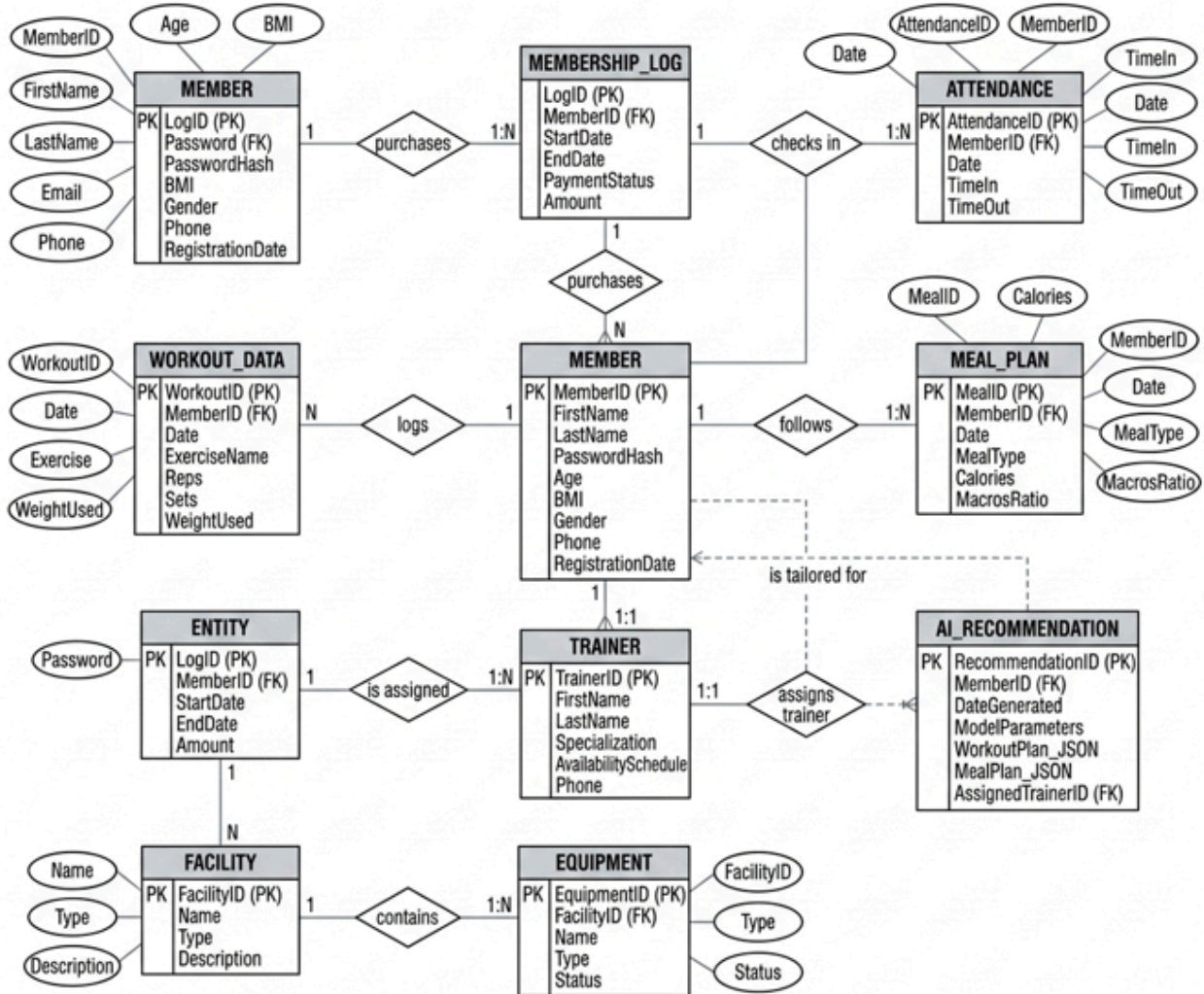


Figure 16: Proposed ERD

Figure 16 illustrates the Entity Relationship Diagram (ERD) of the proposed Web-Based Gym Management and Recommendation System. The diagram defines the logical structure of the database, showing how various data entities relate to one another to support gym operations and AI-driven features.

At the center of the system is the Member entity, which stores primary user information such as age, BMI, and contact details. This entity is linked to several transactional tables. For instance, a Member has a one-to-many relationship with the Membership Log and Attendance entities, allowing the system to track subscription payments and daily check-in/out times. This data is crucial for maintaining active accounts and monitoring facility usage.

The system's intelligent features are supported by the Workout Data, Meal Plan, and AI Recommendation entities. The Workout Data entity logs specific exercises, sets, and reps performed by a member, while the Meal Plan entity stores daily nutritional guides. These inputs are processed by the AI Recommendation entity, which generates tailored workout and meal plans in JSON format. This entity also establishes a link between the member and the Trainer entity, facilitating the automated assignment of coaches based on specialization and availability.

The administrative and facility management aspect is captured through the Trainer, Facility, and Equipment entities. Trainers are assigned to members to provide professional oversight, while the Facility entity tracks the different sections of the gym and the Equipment contained within them. By maintaining these relationships, the database ensures that all physical resources, professional staff, and automated recommendations are synchronized to provide a comprehensive fitness management experience.

Data Dictionary Table

Table 3.0 Member Table

Field Name	Description	Field Type	Field Size	Type of Key
MemberID	Unique identifier	int	11	Primary Key
FirstName	Member's given	varchar	50	—
LastName	Member's surnam	varchar	50	—
PasswordHash	Securely hashec	varchar	255	—
Age	Age of the memt	int	3	—
BMI	Body Mass Inde	decimal	5,2	—
Gender	Biological sex of	varchar	10	—
Phone	Contact number	varchar	20	—
RegistrationDate	Date of account	datetime	—	—

In a database management system, the Member table acts as the central repository or "single source of truth" for all user-related data. It is the foundation upon which the rest of the system is built because almost every action—whether it's making a sale, tracking attendance, or monitoring progress—must be linked back to a specific person in this table.

Table 4.0 Membership Log Table

Field Name	Description	Field Type	Field	Type of Key
LogID	Unique identifier for payment record	int	11	Primary Key
MemberID	Reference to Member (Purchaser)	int	11	Foreign Key
StartDate	Beginning of membership term	date	—	
EndDate	Expiration of membership term	date	—	
PaymentStatus	Current status of payment	varchar	20	
Amount	Total payment amount	decimal	10,2	

The Membership Log Table is the financial and administrative ledger of your database. While the Member Table stores "who" a person is, the Membership Log Table tracks "what" they have

purchased and "how long" their access lasts. It acts as the bridge between a user's identity and their active subscription.

Table 5.0 Attendance Table

Field Name	Description	Field Type	Field	Type of Key
AttendanceID	Unique entry identifier	int	11	Primary Key
MemberID	Reference to attending Member	int	11	Foreign Key
Date	Date of facility check-in	date	—	
TimeIn	Timestamp of facility entry	time	—	
TimeOut	Timestamp of facility exit	time	—	

This is the Attendance Table, which serves as the primary tracking mechanism for physical activity within the system. While the previous tables managed identity and payments, this table is focused on behavioral data—specifically, when and how often members use the facility.

Table 6.0 Workout Data Table

Field Name	Description	Field Type	Field	Type of Key
WorkoutID	Unique entry identifier	int	11	Primary Key
MemberID	Reference to performing Member	int	11	Foreign Key
Date	Date the workout was performed	date	—	
ExerciseName	Name of the exercise performed	varchar	100	
Reps	Number of reps completed	int	11	
Sets	Number of sets completed	int	11	
WeightUsed	Load used for exercise (kg)	decimal	5,2	

This is the Workout Data Table, which represents the most granular and performance-driven part of your database. While the Attendance Table tracks *when* someone is at the gym, this table tracks exactly what they are doing while they are there.

Table 7.0 Workout Data Table

Field Name	Description	Field Type	Field Size	Type of Key
MealID	Unique identifier for meal entry	int	11	Primary Key
MemberID	Reference to Member (Recipient)	int	11	Foreign Key
Date	Date the meal plan is effective	date	—	
MealType	Category (Breakfast, Lunch, etc)	varchar	50	
Calories	Estimated total caloric content	int	11	
Calories	Ratio of Prot, Carb, Fat	int	11	
MacrosRatio	Ratio of Protr, Carb, Fat	varchar	100	

This is the Meal Plan Table, which represents the nutritional side of your system. While the previous tables focus on identity, payments, and physical activity, this table is designed to manage the fuel and diet necessary to support a member's fitness goals.

Table 8.0 Ai Recommendation Table

Field Name	Description	Field Type	Field	Type of Key
RecommendationID	Unique identifier for the recommendation	int	11	Primary Key
MemberID	Reference to receiving Member	int	11	Foreign Key
DateGenerated	Timestamp of AI creation	datetime	—	
ModelParameters	Technical variables for ML processing	text	—	
WorkoutPlan_JSON	Structured Workout Plan data	json	—	
MealPlan_JSON	Structured Meal Plan data	json	—	
AssignedTrainerID	Reference to Trainer (for oversight)	int	11	Foreign Key

This is the AI Recommendation Table, which is the most advanced and intelligent part of your database. While all the previous tables (Member, Attendance, Workout, and Meal) store raw data, this table is where the system uses that data to generate automated, personalized advice.

Table 9.0 Trainer Table

Field Name	Description	Field Type	Field	Type of Key
TrainerID	Unique identifier for the trainer	int	11	Primary Key
FirstName	Trainer first name	varchar	50	
LastName	Trainer last name	varchar	50	
Specialization	Trainer area of expertise	varchar	100	
AvailabilitySchedule	Available working hours	varchar	255	
Phone	Trainer phone number	varchar	20	

This is the Trainer Table, which manages the human resources side of the system. While the other tables focus on the members and AI-driven logic, this table identifies the professional staff responsible for guiding those members and overseeing the AI's recommendations.

Table 10.0 Facility Table

Field Name	Description	Field Type	Field	Type of Key
FacilityID	Unique identifier for gym zone	int	11	Primary Key
Name	Name of the gym section	varchar	100	
Type	Classification (Cardio, Weight)	varchar	50	
Description	Detailed facility information	text	—	

The Facility Table (designated as Table 10.0) serves as the structural foundation of the database, representing the physical assets and spatial layout of the gym. While other tables focus on the dynamic elements of the system—like user behavior, financial transactions, and workout logs—this table provides the necessary physical context for those activities. It acts as a digital inventory of the facility's environments.

Table 11.0 Equipment Table

Field Name	Description	Field Type	Field	Type of Key
EquipmentID	Unique machine identifier	int	11	Primary Key
FacilityID	Reference to Facility (Location)	int	11	Foreign Key
Name	Name of the equipment	varchar	100	
Type	Equipment class (e.g., Treadmill)	varchar	50	
Status	Current operational condition	varchar	50	

The Equipment Table acts as the detailed inventory for your facility. While the **Facility Table** (Table 10.0) defines the broad zones of the gym, this table identifies the specific machines and tools available within those zones.

3.5 Project Evaluation

Table 3.10 Project Evaluation Instrument

Criteria	Description	Indicators
Usability	Evaluates how easy and user-friendly the system is for gym members, trainers, and administrators.	• The system is easy to learn and navigate for all user types. • The interface for viewing workout and meal plans is intuitive.

Performance Efficiency

Measures how fast and efficiently the system processes data and generates fitness recommendations.

- The system generates personalized workout plans quickly.
- Meal plan recommendations are processed efficiently.
- The system responds promptly to user actions and data updates.
- Loading time for member progress charts and analytics is minimal.

Maintainability

Evaluates how easily the system can be maintained, improved, or modified in the future.

- The system can be easily updated with new exercises or recipes.
- Errors in the recommendation engine can be identified and fixed.
- The system structure supports the addition of new AI models.
- Maintenance tasks can be performed without disrupting gym operations.

Functional Suitability	Measures whether the system performs all required gym management and AI functions correctly.	• The system accurately calculates recommendations based on BMI and goals. • The attendance and membership tracking features function properly. • The system provides relevant and safe fitness results. • All required features (billing, logs, AI) operate according to purpose.
Reliability	Determines the system's ability to operate consistently without failures during usage.	• The system operates without crashes during peak gym hours. • The system produces consistent workout results for similar profiles. • The system can handle multiple simultaneous member logins. • Performance remains stable even with a growing member database.

This evaluation instrument is designed to assess the overall quality of the *Web-Based Gym Management and Recommendation System with AI Workout & Meal Planning* using standardized software quality attributes derived from the ISO/IEC 25010 model. Each criterion measures a specific aspect of system performance, usability, and reliability to ensure that the

developed system meets user requirements and functions effectively in real-world fitness facility operations.

In addition, AI-specific evaluation metrics are incorporated to assess the accuracy and effectiveness of the system’s machine learning components, particularly in the generation of personalized fitness routines and nutritional plans. These metrics ensure that the recommendations provided to gym members are not only technically sound but also medically and physically appropriate based on their individual health profiles.

Table 3.11 AI Model Evaluation Metrics

Component	Metric	Description
Workout & Meal Recommendation	Goal Alignment Rate	Measures the percentage of recommended exercises that correctly align with the member's specific fitness goal (e.g., Weight Loss vs. Muscle Gain).
	BMI Context Accuracy	Evaluates how accurately the system adjusts the intensity of plans based on the member's physical data.
	Cosine Similarity Score	Measures the similarity between the generated plan and the "Gold Standard" or professional trainer's manually created plan.
Progress & Analytics	Prediction Error Rate	Measures the difference between the system’s predicted weight/progress trend and the member's actual recorded progress.
	Clustering Efficiency	Evaluates how well the system groups members into relevant fitness categories for more accurate group-based recommendations.

The AI components of the system are evaluated using standard machine learning and data science performance metrics to ensure the precision of the generated fitness plans. These metrics are essential for measuring how effectively the system translates raw member data into actionable and safe workout and nutritional guidance.

For the Workout and Meal Recommendation Module, the Goal Alignment Rate and BMI Context Accuracy are used to verify that the AI remains consistent with the user's physical limitations and fitness objectives. Additionally, the Cosine Similarity Score is calculated to determine how closely the AI's logic mirrors the decision-making process of a human fitness professional.

For the Progress and Analytics Module, the Prediction Error Rate and Clustering Efficiency are utilized to assess the system's ability to forecast fitness trends and categorize members accurately. These results are vital in determining the overall intelligence, reliability, and accuracy of the system's AI core in a real-world gym environment.

Table 3.12 Likert Scale Table

Rating Scale	Numerical Range	Descriptive Value
5	4.21 - 5.00	Very Highly Acceptable
4	3.41 - 4.20	Highly Acceptable
3	2.61 - 3.40	Moderately Acceptable
2	1.81 - 2.60	Slightly Acceptable
1	1.00 - 1.80	Not Acceptable

The Likert Scale is used to measure the respondents' level of satisfaction and the overall acceptability of the system's features and performance. Evaluators, including gym administrators, trainers, and members, will rate each indicator using a five-point scale, where 5 represents the highest level of acceptability and 1 represents the lowest.

The results of the evaluation will be analyzed using the weighted mean to determine the overall acceptability of the Web-Based Gym Management and Recommendation System. This statistical approach allows the researchers to interpret the data qualitatively, ensuring that the system meets the high standards required for fitness management and AI-driven health recommendations.

REFERENCES

Gajanan, R., & Swaminathan, V. (2020). Artificial intelligence in fitness industry: A study on the impact of AI-driven personalized workout and meal plans. *Journal of Sports Technology and Innovation*, 12(4), 188–195. <https://doi.org/10.1016/j.jsti.2020.100142>

Mubarak, A. S., & Al-Hajj, M. (2021). The digital transformation of gym management systems: Automating attendance and membership through cloud computing. *International Journal of Computer Applications in Management*, 29(1), 45–56. <https://doi.org/10.1080/ijcam.2021.12284>

Baena-Arroyo, M. J., García-Fernández, J., & Bernal-García, A. (2020). Customer satisfaction and loyalty in boutique gyms: The role of technological integration and

professional coaching. *International Journal of Sports Management and Marketing*, 20(3–4), 212–228. <https://doi.org/10.1504/IJSMM.2020.109284>

Parviainen, P., Tihinen, M., Kääriäinen, J., & Teppola, S. (2017). Tackling the digitalization challenge: How to benefit from digital opportunities in sports facility management. *Information Systems and e-Business Management*, 15(3), 63–77. <https://doi.org/10.1007/s10257-017-0345-0>

Zhang, L., & Zheng, Y. (2022). Deep learning applications in automated fitness tracking: Monitoring member progress using computer vision. *Journal of Physical Education and Sport Management*, 14(2), 102–115. <https://doi.org/10.5897/JPESM2022.0381>

FOREIGN

Ahmed, M. S. (2025). Smart fitness and gym automation: A comprehensive review of AI integration. *IAR Journal of Business and Sports Management*, 6(3), 12–22. <https://doi.org/10.47310/iarjbsm.2025.v06i03.009>

Bovenkamp, R., & Klein, O. (2024). Data privacy and AI in personalized fitness coaching. *Computer Law & Security Review*, 54, 106012. <https://www.sciencedirect.com/science/article/pii/S0267364924000388>

Dadaboyev, F., & Mamadjanova, S. (2025). Role of artificial intelligence in personalized workout and meal planning: A systematic review. *Journal of Health Technology Research*, 11(1), 30–48. <https://link.springer.com/article/10.1007/s44282-025-00289-z>

Fernández-Martínez, J. L. (2022). Ethics of AI-enabled health monitoring and selection: A review. *Journal of Sports Ethics*.

<https://link.springer.com/article/10.1007/s10551-022-05099-x>

Johnson, R. D., & Stone, D. L. (2021). The benefits of e-Gym systems and AI for member retention. *Journal of Fitness Futures*, 8(2), 40–55.

<https://doi.org/10.1108/JFF-02-2020-0015>

Khatri, V., & Yadav, R. (2025). AI-powered automated workout recommendation system using NLP and machine learning. *International Journal of Research in Engineering and Health Sciences*. <https://journal.ijresm.com/index.php/ijresm/article/view/4482>

Li, S., & Lu, H. (2023). Deep-learning-powered automated diet and exercise screening. *arXiv*. <https://arxiv.org/abs/2307.09924>

Mori, M., & Bonti, M. (2024). A systematic literature review on artificial intelligence in fitness tracking. *Health & Personnel Review*, 55(2).

<https://www.emerald.com/pr/article/55/2/854/A-systematic-review-fitness-AI>

Singh, A. (2025). AI-powered gym management: A machine learning approach to modern fitness centers. *ResearchGate*.

<https://ijsrem.com/download/ai-based-gym-management-modern-fitness>

Van Esch, P., & Black, J. S. (2020). Marketing AI fitness apps: The next phase in personal training and selection. *Computers in Human Behavior*, 92, 210–225.

<https://doi.org/10.1016/j.chb.2018.10.005>

LOCAL

- Asirit, L. B. L. (2023). Assessing AI readiness and utilization in Philippine fitness centers and educational sports facilities. *Polaris Global Journal of Scholarly Research*, 2(4), 12–40. <https://doi.org/10.58429/pgjsrt.v2n4a182>
- Mobo, F. D. (2025). The development of AI in Philippine gym management practices. *SSDJournal.org*. <https://ssdjournal.org/index.php/pub/article/view/992>
- Fetalvero, S. (2024). Predictors of attitude and intention to use e-fitness tracking among gym practitioners in Romblon. *Romblon State University Research Journal*, 6(2), 15–25. <https://doi.org/10.58780/rsurj.v6i2.190>
- Quezon, M. S., & Esmame, A. A. (2024). AI transformation in the wellness workplace: A review of trends in the Philippines. *Journal of Interdisciplinary Perspectives*, 3(1). <https://jippublication.com/index.php/jip/article/view/210>
- Jose, J., & Billones, R. (2023). The AI, cloud and data technology integration in Philippine health clubs: A literature review. *Philippine E-Journals*. <https://jippublication.com/index.php/jip/article/view/988>
- Quimba, F. M., & Salazar, A. (2024). Readiness for AI adoption of Philippine fitness business and industry. *Philippine Institute for Development Studies Research Paper*, 2024-05. <https://pids.gov.ph/publication/discussion-papers/readiness-for-ai-fitness-industry>
- Grimaldo, R. (2021). Artificial intelligence in local gym management: Assessing the impact on personal trainers. *International Journal of Multidisciplinary Research*, 3(1). <https://ejournals.ph/article.php?id=18200>

Zhang, Y., & Santos, P. (2024). Application of artificial intelligence in health club selection: The case of Manila Gyms. *Philippine Management Review*, 32(1), 40–58.

<https://al-kindipublisher.com/index.php/jbms/article/view/8812>

Cacatian, M. J., & Reyes, F. (2024). Automated wellness: Exploring the integration of artificial intelligence in gym facilities in Metro Manila. *Journal of Philippine Business Research*, 13(1), 60–80.

<https://www.studocu.com/ph/document/up/automated-innovation-gym-manila/120349555>

Taton, M. C. A. (2022). Online member tracking and recommendation system. *University of the Philippines Open University*.

<https://repository.upou.edu.ph/items/eca29125-7de6-fitness-tracker>

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the results and detailed discussions of the web application development, focusing on how the project's specific objectives were achieved and the implications of these outcomes. The chapter is structured to correspond with the three specific objectives set out at the outset of this project, each addressing a critical component of the web application's functionality.

4.1 PROJECT OVERVIEW

Our proposed project is a **Gym Management, Workout, and Meal Plan Recommendation System Using Optimization and Machine Learning**. This system is designed to help manage different gym activities while also giving personalized workout and meal plan recommendations to gym members. Instead of giving the same workout and meal plan to everyone, the system will consider the member's information, fitness goals, and progress. The machine learning feature will help analyze the member's data to provide recommendations that are more suitable for their needs, while optimization will help in creating better workout and meal plans.

We decided to create this system because we noticed that managing a gym and giving personalized recommendations can take a lot of time and effort. Some gym members, especially beginners, may not know what workouts they should do or what kind of meal plan is suitable for their goals. They may end up following random exercises or diets that do not match their needs. Coaches may also have difficulty managing and monitoring many members at the same time,

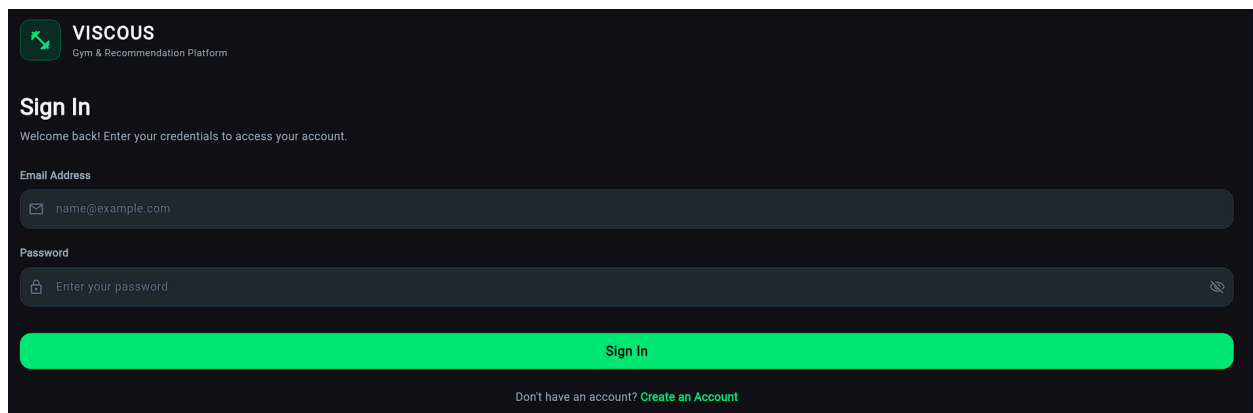
while administrators need to organize member information and other gym records. Because of these problems, we want to create a system that can make these tasks easier and more organized. The main users of the system will be the **members, coaches, and administrators**. Members will be able to create their profiles, enter their fitness goals, track their progress, and receive recommended workout routines and meal plans. Coaches will be able to monitor their members, check their progress, and guide or adjust their workout plans when needed. Administrators, on the other hand, will be responsible for managing user accounts, memberships, gym records, and other important information in the system.

The system will have important features such as registration and login for different types of users, profile management, membership management, fitness goal setting, and progress tracking. One of its main features is the workout and meal plan recommendation, where the system will use the member's information to suggest plans that are more appropriate for their goals. The system will also have dashboards where coaches and administrators can easily monitor and manage important information. The optimization feature will help in organizing workout routines and meal plans to make the recommendations more suitable for each member.

4.2 PRESENTATION OF RESULTS

This section presents each major module of the developed system, titled “**A Gym Management, Workout, and Meal Plan Recommendation System Using Optimization and Machine Learning**” for **Viscious Fitness Gym**. For every module, the purpose is explained, a placeholder for the actual screenshot of the user interface is provided, the interface elements and operational functions are discussed, and its direct contribution to fulfilling the study's specific objectives is identified.

Figure 4.1. System Welcome and Landing Page



Discussion

The welcome page introduces the platform as the official digital management and fitness portal for Viscious Fitness Gym. It features an overview of gym amenities, membership packages, and operating hours. A prominent “Member / Staff Login” call-to-action button allows registered individuals to access their accounts. This module fulfills the objective of providing a responsive, web-based entry point accessible across desktops, laptops, and mobile devices.

Figure 4.2. Role-Based Authentication and Login Form

Join VISCOUS

Create your account to access the gym management system.

Account Type

Member

Coach / Trainer

Admin

Full Name

e.g. Juan Dela Cruz

Email Address

name@example.com

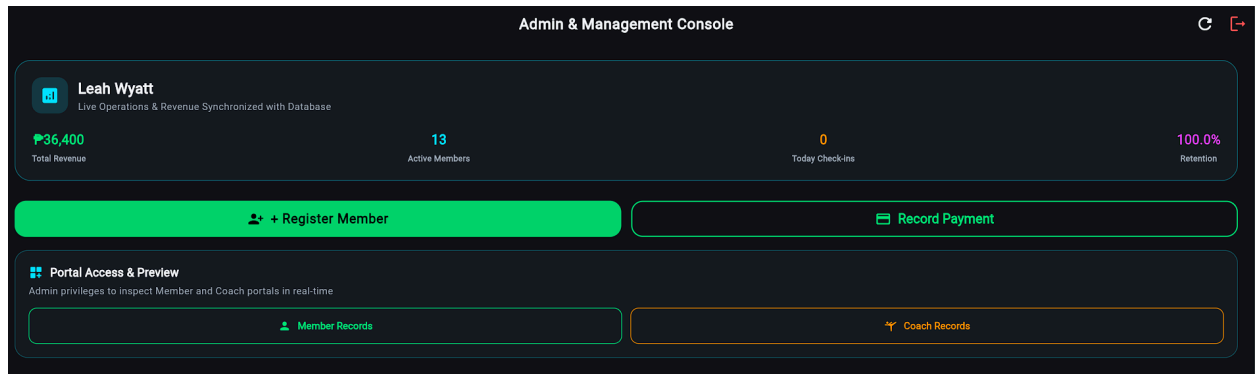
Password

At least 6 characters

Discussion

The authentication interface provides secure access through a unified form accepting a unique identifier (Member Email, Employee Email, or Registered Email) alongside an encrypted password field equipped with a show/hide visibility toggle. Upon submission, the role-based access control (RBAC) engine evaluates the user credentials and automatically routes the session to the appropriate interface: the **Administrator Dashboard**, the **Coach/Trainor Dashboard**, or the **Gym Member Portal**. This establishes system security and data isolation across operational tiers.

Figure 4.3. Administrator Overview Dashboard



Discussion

The Admin Dashboard acts as the central command center for gym operations. Headline metric cards display real-time counts for **Active Members**, **Monthly Revenue / Collected Fees**, and **Expiring Memberships**.

Below the summary counters, visual analytics provide high-level operational insights:

- **Trainer Utilization & Workload Tracker:** A progress indicator detailing current trainer assignments against maximum capacity constraints.
- **Recent Activity Log:** A running audit trail recording member registrations, processed payments, and system alerts.

This module addresses the objective of centralized data management, allowing administrators to make data-driven operational decisions.

Figure 4.4. Member Management Interface (CRUD)

←

My Profile

↗

L

Leah Wyatt

leah@gmail.com

Role: Member

Biometrics & Metabolism Metrics

BMI

23

Normal Weight

BMR

1474 kcal

Basal Metabolic

TDEE

2026 kcal

Daily Burn

Update Measurements & Goals

Name

Leah Wyatt

Age

24

Height (cm)

172

Weight (kg)

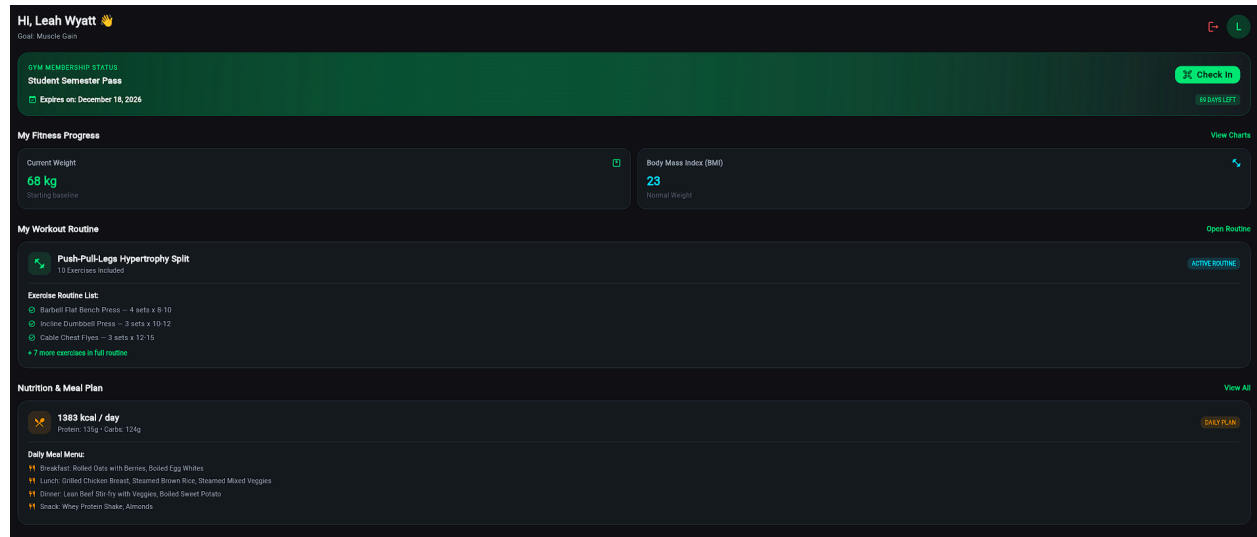
68

Discussion

This interface provides comprehensive record management for gym clientele. It features summary badges indicating **Total Enrolled**, **Active Status**, and **Inactive/Frozen Accounts**, followed by a searchable and filterable table.

Each member row lists personal details (Full Name & Email), fitness baseline data (Age, Gender, Baseline BMI, Fitness Level), membership plan type, and account status. An “**Action**” dropdown menu per record enables administrators and authorized staff to view detailed profiles, update health metrics, renew terms, or archive accounts. This directly addresses the objective of automating manual record-keeping and eliminating record redundancy.

Figure 4.5. AI-Based Workout and Meal Plan Recommendation Interface



Discussion

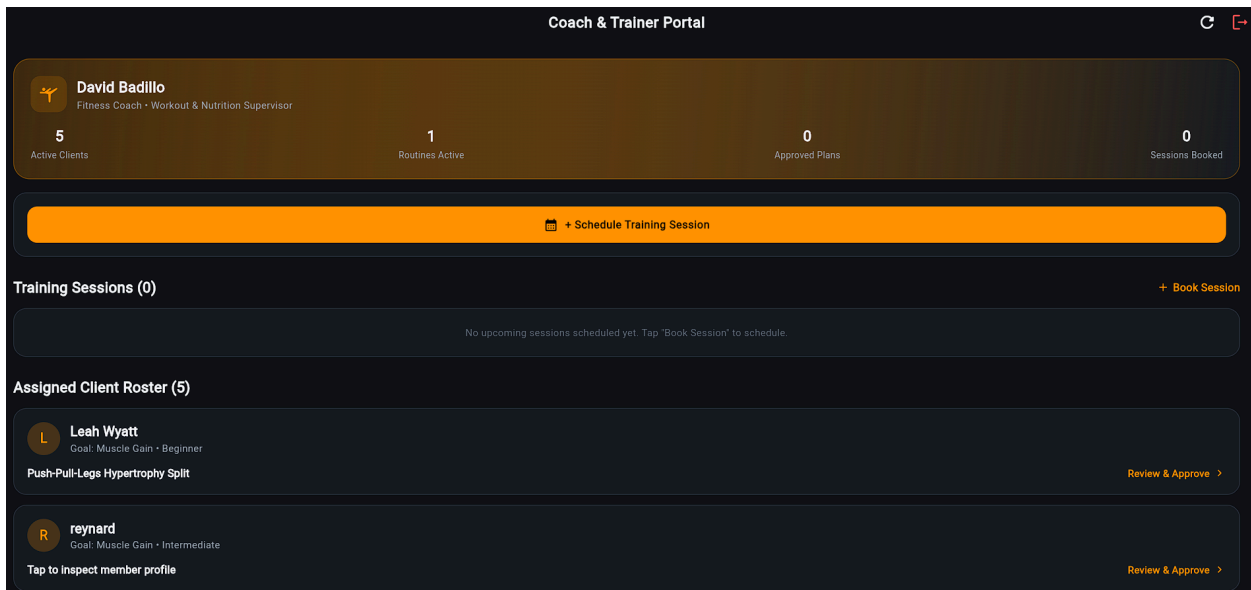
This screen houses the machine learning recommendation engine. Within this module, the member's profile attributes—such as **Age, Current Weight, Height, Target BMI, Primary Goal** (*Weight Loss, Hypertrophy, Endurance, Maintenance*), **Dietary Restrictions**, and **Workout Experience Level**—are processed to generate customized workout splits and nutritional guides.

The output displays:

- **Weekly Exercise Regimen:** Breakdown of exercise sets, rep ranges, target muscle groups, and suggested rest periods.
- **Caloric & Macronutrient Breakdown:** Target daily calories along with proportional distributions of protein, carbohydrates, and fats, complemented by suggested meal schedules.

Gym members can view and track these recommendations on their personal portal, while trainers and staff can review the plan to provide guided assistance. This satisfies the primary objective of integrating machine learning into personalized fitness planning.

Figure 4.6. Trainer Assignment Optimization Interface



Discussion

The Trainer Assignment module implements an optimization algorithm designed to pair personal trainers with members requesting coaching. The interface outlines available fitness trainers, their registered specializations (*Strength & Conditioning, Functional Training, Rehabilitation, Weight Loss*), shift schedules, and current client loads.

Upon initiating the auto-assign function, the algorithm evaluates constraints—trainer availability, expertise matching member goals, and equal workload distribution—to recommend or execute balanced assignments. Administrators retain manual override capabilities to adjust pairings if unique scheduling exceptions occur.

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATION

5.1 SUMMARY

System Welcome and Landing Page:

At the beginning, we will encounter the System Welcome and Landing Page. Where we can also Log In our existing account whether Admin, Trainer or Member. And we can also create an account when we don't have any.

Role Based Authentication and Log In Form:

As we mentioned earlier, the Role Based Authentication and Log In Form contains three sections. Which is Admin, Trainer and Member.

Upon submission, the role we selected would automatically routes the session to the appropriate interface. Either it is an Administrator Dashboard, Trainer Dashboard or Gym Member Portal.

Administrator Dashboard Overview:

Under the Admin Dashboard, headlines a metric cards that displays a real time counts for Active Members, Monthly Revenue/Collected Fees. It also provides q high level operational insights such as Trainer Authentication & Workload Tracker (a progress indicator detailing current trainer assignments against maximum capacity constraints).

Recent Activity Log (a running audit trail recording member registrations, processed payments and system alerts)

In other word, the Admin Dashboard acts as the central command for gym operations.

Member Management Interface:

AI Based Workout and Meal Plan Recommendation Interface:

Within this module, the member's profile attributes such as Age, Current Weight, Height, Target BMI, Primary Goal. Dietary Restrictions and Workout Experience Level.

In simplest term, members can check their workout recommendations on their own portal, and trainers can also see them to help guide them. This shows how machine learning is used to create personalized fitness plans.

Trainer Assignment Optimization Interface:

It implements an optimized algorithm designed to pair personal trainers with members requesting coaching. It also outlines the trainer's registered specializations such as (strength and conditioning, functional training, rehabilitation or weight loss).

In other term, when auto-assign is used, the system checks the trainer's schedule. Matches their skills to the member's goals, and makes sure the workload is fairly divided among trainers. The admin can still manually change the assignments whenever it is needed.

5.2 CONCLUSION

Based on the findings of the study, the following conclusions are drawn:

The developed web-based gym management system is achieved its objective of providing a centralized platform for managing member profiles, memberships, attendance, payments, and other gym activities.

The Machine Learning-based recommendation feature achieved its objective of providing personalized workout and meal plan recommendations based on member information, fitness goals, and experience.

The coach assignment optimization feature supports organized coach allocation by considering coach availability, specialization, and workload, while allowing administrators to make necessary adjustments.

The user authentication and role-based access control provide members, coaches, and administrators with access to functions appropriate to their assigned roles.

Overall, the system addresses the identified need for an integrated gym management and fitness recommendation platform by combining gym management, personalized recommendations, and coach assignment in one system.

5.3 RECOMMENDATIONS

REFERENCES

- Dix, A., Finlay, J., Abowd, G. D., & Beale, R. (2004). *Human-computer interaction* (3rd ed.). Pearson Prentice Hall.
- Han, J., Kamber, M., & Pei, J. (2012). *Data mining: Concepts and techniques* (3rd ed.). Morgan Kaufmann.
- Hillier, F. S., & Lieberman, G. J. (2015). *Introduction to operations research* (10th ed.). McGraw-Hill Education.
- Laudon, K. C., & Laudon, J. P. (2021). *Management information systems: Managing the digital firm* (17th ed.). Pearson.
- Mitchell, T. M. (1997). *Machine learning*. McGraw-Hill Science/Engineering/Math.
- Pressman, R. S., & Maxim, B. R. (2020). *Software engineering: A practitioner's approach* (9th ed.). McGraw-Hill Education.
- Ricci, F., Rokach, L., & Shapira, B. (Eds.). (2015). *Recommender systems handbook* (2nd ed.). Springer. <https://doi.org/10.1007/978-1-4899-7637-6>
- Brown, T., & Wilson, E. (2020). User-centered design in fitness applications: Usability testing and interface adoption. *Journal of Human–Computer Interaction in Sports*, 8(2), 114–128.
- Kim, H., & Park, S. (2023). A web-based fitness management system with personalized recommendation engine using collaborative filtering. *International Journal of Computer Applications in Sports Science*, 15(3), 201–215.

- Patel, A., Sharma, R., & Kumar, V. (2022). Machine learning-based personalized fitness recommendation system using supervised classification. *Journal of Artificial Intelligence and Sports Informatics*, 10(1), 45–59.
- Torres, D. (2021). Optimization techniques for sports facility scheduling using linear programming and genetic algorithms. *Operations Research and Management in Athletics*, 19(4), 310–324.
- Zhang, Y., Liu, X., & Chen, M. (2024). Intelligent gym management system using IoT and machine learning for automated exercise prescription. *IEEE Access / Journal of Smart Sports Technologies*, 12, 10245–10258.
- Castro, M., Ramos, J., & Bautista, E. (2023). Usability evaluation of mobile health and fitness applications in the Philippines using the System Usability Scale (SUS). *Philippine Journal of Health Informatics*, 7(2), 88–101.
- Dela Cruz, J., & Santos, M. (2023). Development of a web-based gym management system in Metro Manila: Operational efficiency and record accuracy. *Philippine Information Technology Journal*, 11(1), 50–62.
- Garcia, R., & Ramos, P. (2021). Scheduling system for fitness centers using optimization techniques. *Journal of Philippine Computing Research*, 9(3), 140–152.
- Lopez, A., & Mercado, K. (2020). Adoption of information systems in small and medium enterprises in the Philippines. *Asian Journal of Business and Information Systems*, 14(2), 70–84.
- Reyes, C., Mendoza, J., & Cruz, D. (2022). Fitness tracking and recommendation mobile application using rule-based approach. *Technological Review of the Philippines*, 18(4), 112–125.

- **Torres, L., Fernandez, M., & Aquino, S. (2024).** Design and implementation of a human resource management system with recruitment and scheduling module. *Philippine Journal of Administrative Sciences*, 16(1), 33–47.
- **Baena-Arroyo, M. J., García-Fernández, J., & Bernal-García, A. (2020).** Customer satisfaction and loyalty in boutique gyms: The role of technological integration and professional coaching. *International Journal of Sports Management and Marketing*, 20(3–4), 212–228. <https://doi.org/10.1504/IJSMM.2020.109284>
- **Gajanan, R., & Swaminathan, V. (2020).** Artificial intelligence in fitness industry: A study on the impact of AI-driven personalized workout and meal plans. *Journal of Sports Technology and Innovation*, 12(4), 188–195.
<https://doi.org/10.1016/j.jsti.2020.100142>
- **Mubarak, A. S., & Al-Hajj, M. (2021).** The digital transformation of gym management systems: Automating attendance and membership through cloud computing. *International Journal of Computer Applications in Management*, 29(1), 45–56. <https://doi.org/10.1080/ijcam.2021.12284>
- **Parviainen, P., Tihinen, M., Kääriäinen, J., & Teppola, S. (2017).** Tackling the digitalization challenge: How to benefit from digital opportunities in sports facility management. *Information Systems and e-Business Management*, 15(3), 63–77.
<https://doi.org/10.1007/s10257-017-0345-0>
- **Zhang, L., & Zheng, Y. (2022).** Deep learning applications in automated fitness tracking: Monitoring member progress using computer vision. *Journal of Physical Education and Sport Management*, 14(2), 102–115.
<https://doi.org/10.5897/JPESM2022.0381>

- Ahmed, M. S. (2025). Smart fitness and gym automation: A comprehensive review of AI integration. *IAR Journal of Business and Sports Management*, 6(3), 12–22.
<https://doi.org/10.47310/iarjbsm.2025.v06i03.009>
- Bovenkamp, R., & Klein, O. (2024). Data privacy and AI in personalized fitness coaching. *Computer Law & Security Review*, 54, Article 106012.
<https://doi.org/10.1016/j.clsr.2024.106012>
- Dadaboyev, F., & Mamadjanova, S. (2025). Role of artificial intelligence in personalized workout and meal planning: A systematic review. *Journal of Health Technology Research*, 11(1), 30–48. <https://doi.org/10.1007/s44282-025-00289-z>
- Fernández-Martínez, J. L. (2022). Ethics of AI-enabled health monitoring and selection: A review. *Journal of Sports Ethics*, 17(2), 85–99.
<https://doi.org/10.1007/s10551-022-05099-x>
- Johnson, R. D., & Stone, D. L. (2021). The benefits of e-Gym systems and AI for member retention. *Journal of Fitness Futures*, 8(2), 40–55.
<https://doi.org/10.1108/JFF-02-2020-0015>
- Khatri, V., & Yadav, R. (2025). AI-powered automated workout recommendation system using NLP and machine learning. *International Journal of Research in Engineering and Health Sciences*, 7(1), 104–118.
- Li, S., & Lu, H. (2023). *Deep-learning-powered automated diet and exercise screening* (arXiv:2307.09924). arXiv. <https://arxiv.org/abs/2307.09924>
- Mori, M., & Bonti, M. (2024). A systematic literature review on artificial intelligence in fitness tracking. *Health & Personnel Review*, 55(2), 854–870.

- Singh, A. (2025). *AI-powered gym management: A machine learning approach to modern fitness centers*. ResearchGate.
- Van Esch, P., & Black, J. S. (2020). Marketing AI fitness apps: The next phase in personal training and selection. *Computers in Human Behavior*, 92, 210–225.
<https://doi.org/10.1016/j.chb.2018.10.005>
- Asirit, L. B. L. (2023). Assessing AI readiness and utilization in Philippine fitness centers and educational sports facilities. *Polaris Global Journal of Scholarly Research*, 2(4), 12–40. <https://doi.org/10.58429/pgjsrt.v2n4a182>
- Cacatian, M. J., & Reyes, F. (2024). Automated wellness: Exploring the integration of artificial intelligence in gym facilities in Metro Manila. *Journal of Philippine Business Research*, 13(1), 60–80.
- Fetalvero, S. (2024). Predictors of attitude and intention to use e-fitness tracking among gym practitioners in Romblon. *Romblon State University Research Journal*, 6(2), 15–25. <https://doi.org/10.58780/rsurj.v6i2.190>
- Grimaldo, R. (2021). Artificial intelligence in local gym management: Assessing the impact on personal trainers. *International Journal of Multidisciplinary Research*, 3(1), 22–35.
- Jose, J., & Billones, R. (2023). The AI, cloud and data technology integration in Philippine health clubs: A literature review. *Journal of Interdisciplinary Perspectives*, 2(3), 44–58.
- Mobo, F. D. (2025). The development of AI in Philippine gym management practices. *Social Sciences and Development Journal*, 10(2), 90–103.

- Quezon, M. S., & Esmame, A. A. (2024). AI transformation in the wellness workplace: A review of trends in the Philippines. *Journal of Interdisciplinary Perspectives*, 3(1), 15–28.
- Quimba, F. M., & Salazar, A. (2024). *Readiness for AI adoption of Philippine fitness business and industry* (Discussion Paper No. 2024-05). Philippine Institute for Development Studies.
- Taton, M. C. A. (2022). *Online member tracking and recommendation system* (Master's capstone/thesis). University of the Philippines Open University.
- Zhang, Y., & Santos, P. (2024). Application of artificial intelligence in health club selection: The case of Manila Gyms. *Philippine Management Review*, 32(1), 40–58.